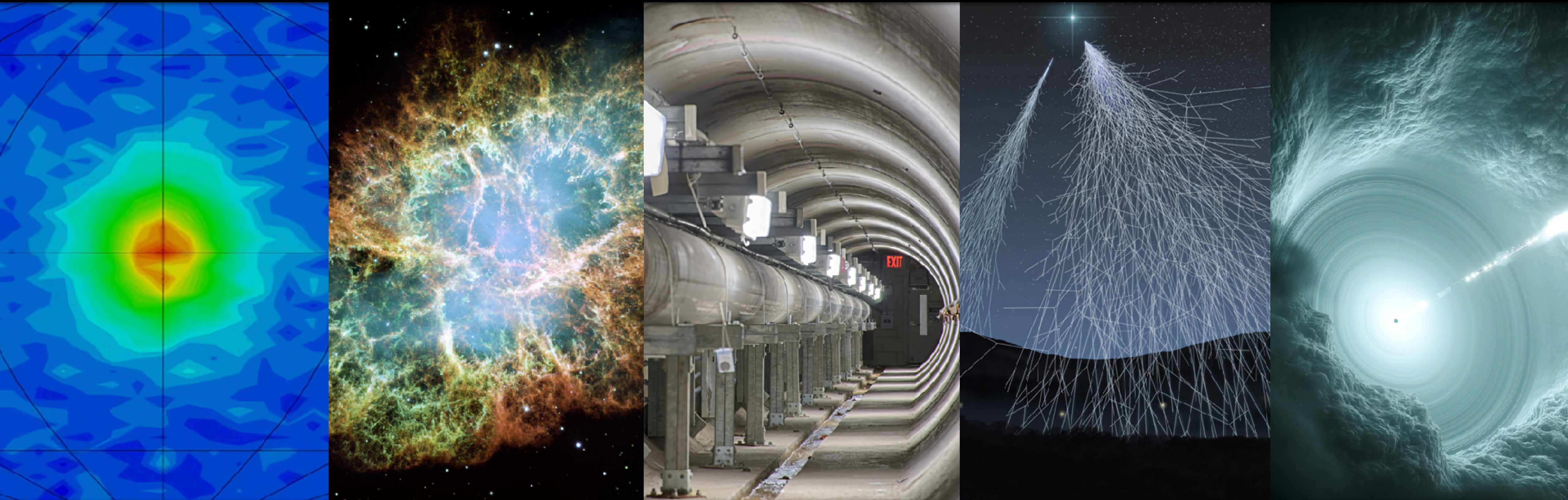


Neutrino Facilities as Versatile Probes of New Physics

APS Global Summit, March 16 2026



Matheus Hostert matheus-hostert@uiowa.edu
University of Iowa

IOWA

1) Neutrinos as unique building blocks in extensions of the Standard Model.

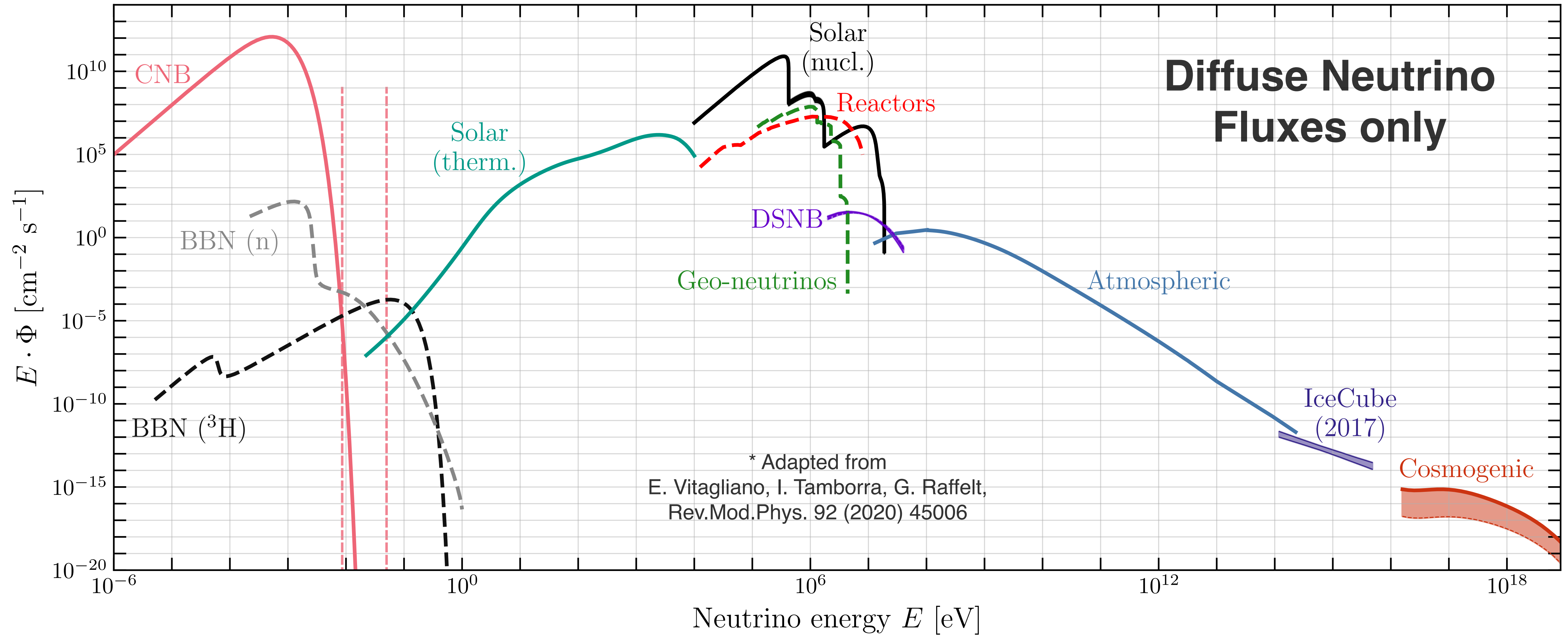
- the only $U(1)_{\text{EM}}$ -singlet fermions of the SM,
- mass mechanism unknown.

2) Neutrino experiments as probes of new particles and forces.

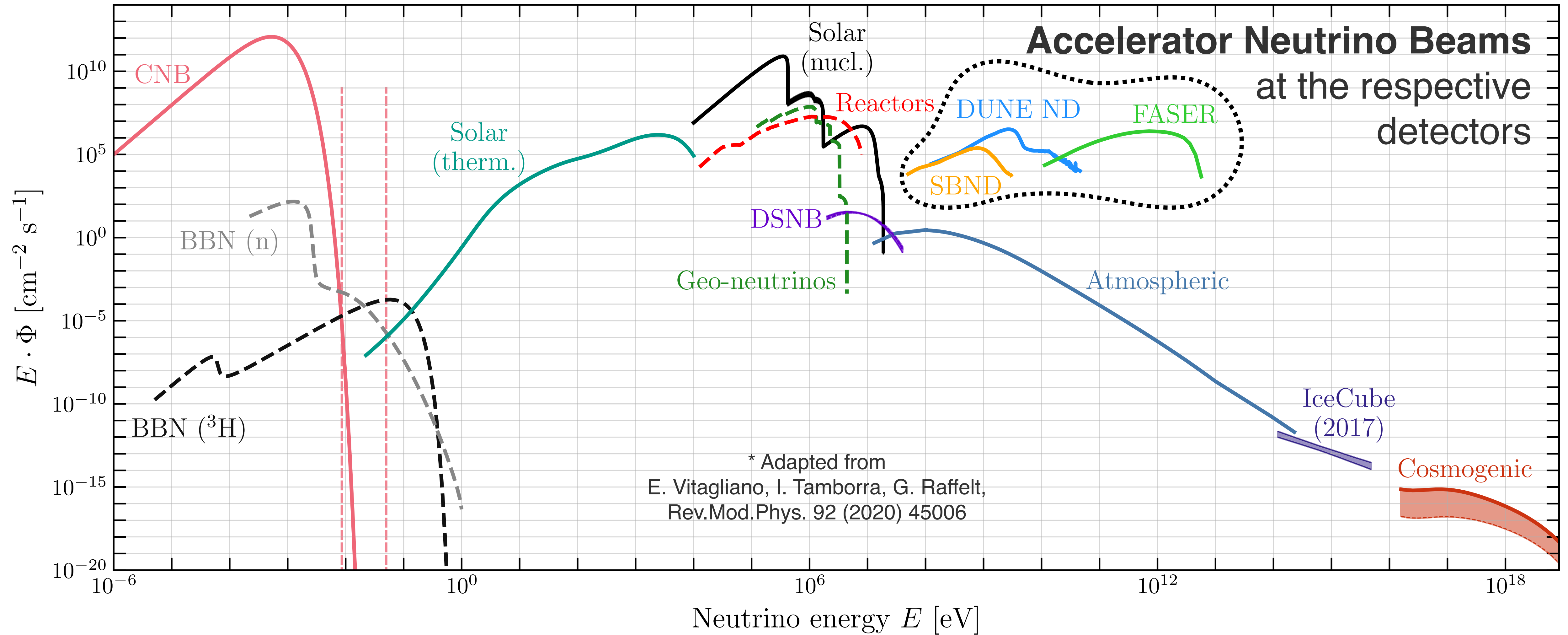
- built for rare phenomena and helping push intensity frontier,
- Several anomalies to be resolved.



Neutrinos Across Energy Scale

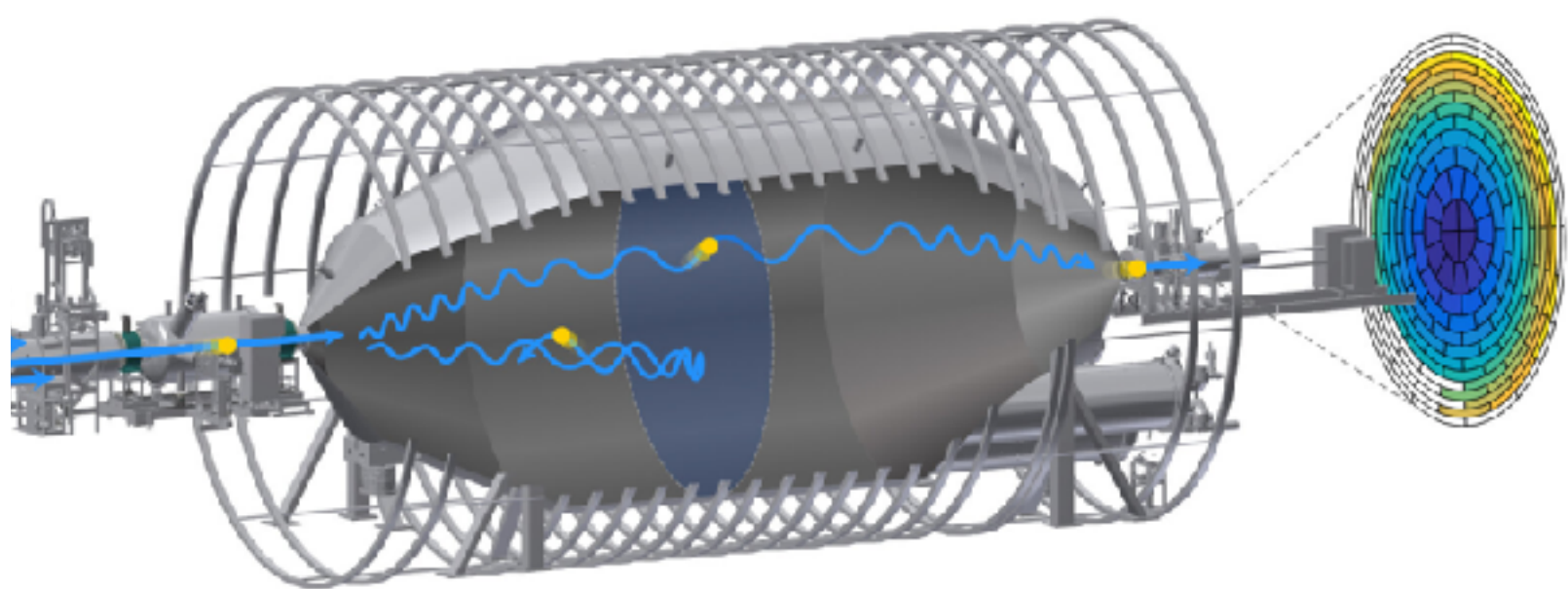


Neutrinos Across Energy Scale

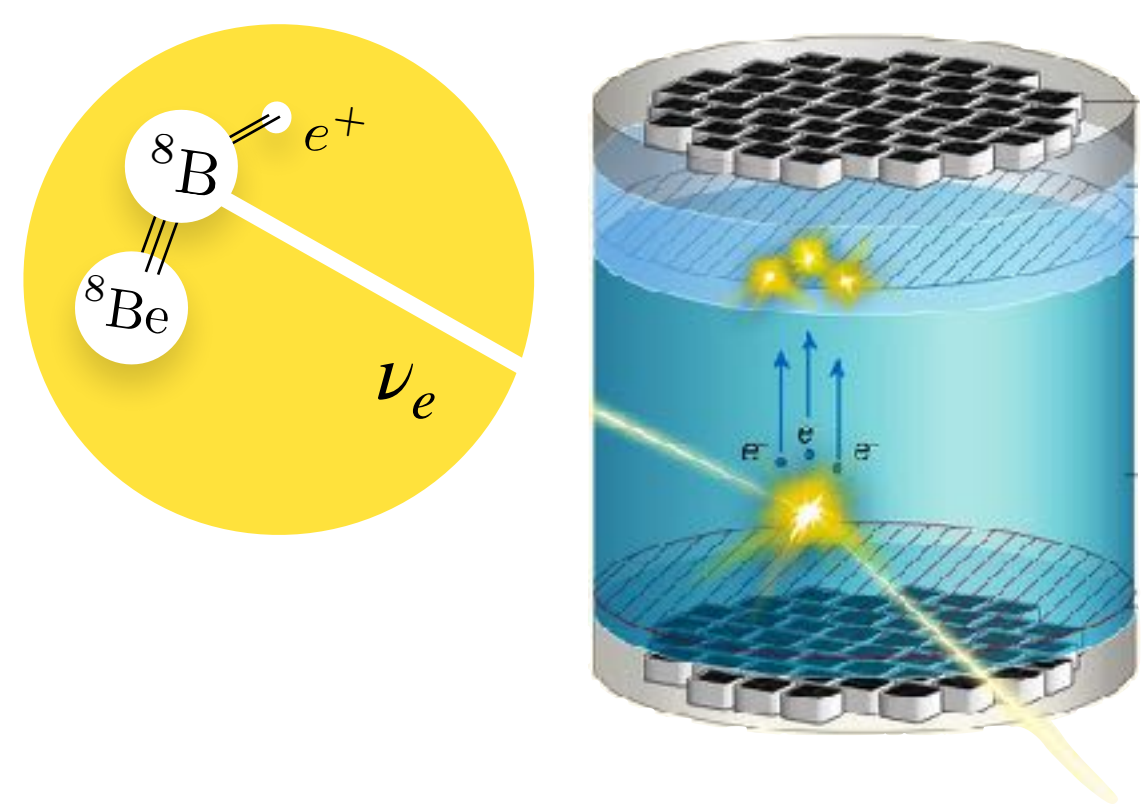


Near future: covering MeV $\rightarrow$ EeV energies with state-of-the-art measurements

Beta decay spectra

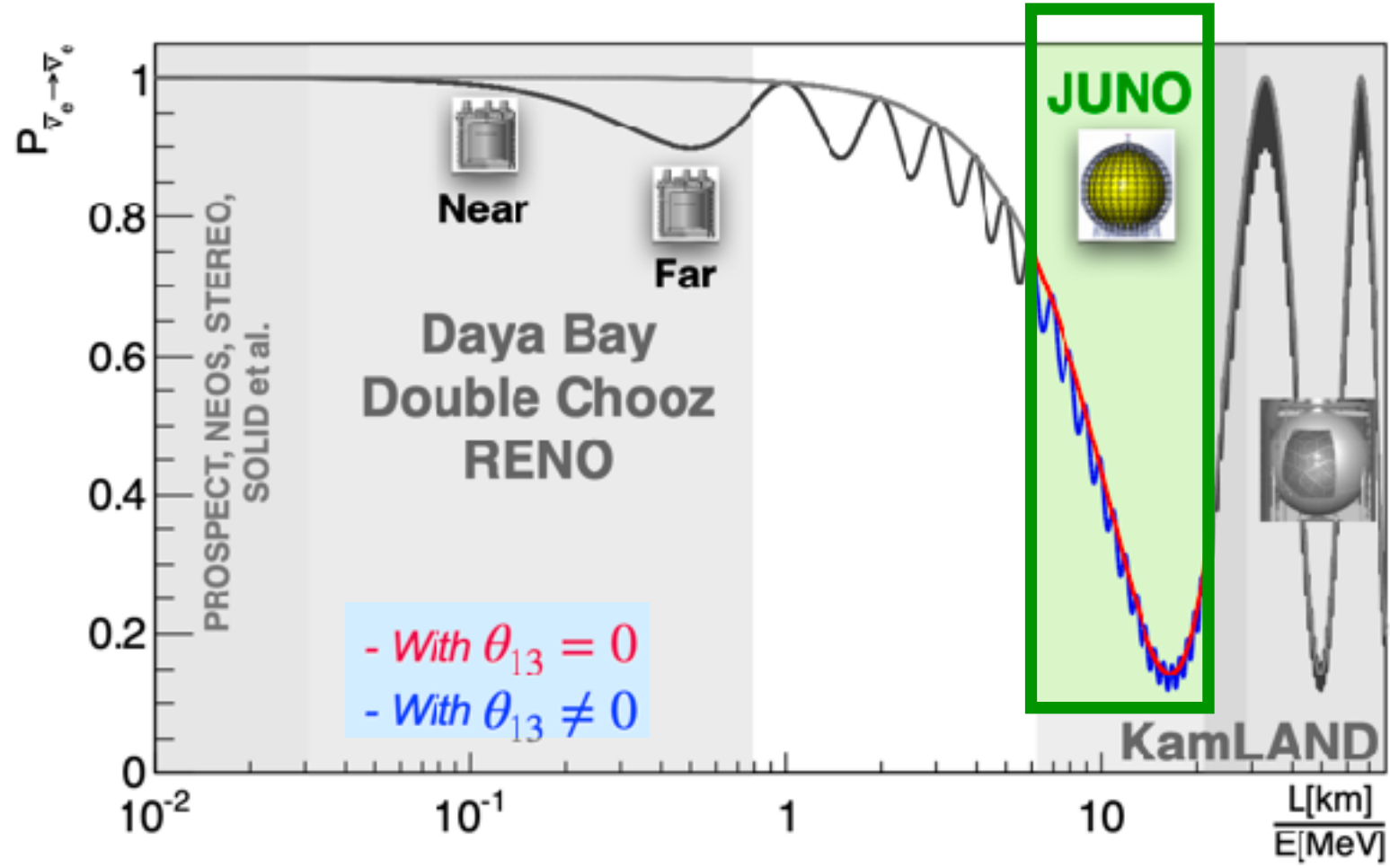


Dark Matter Direct Detection

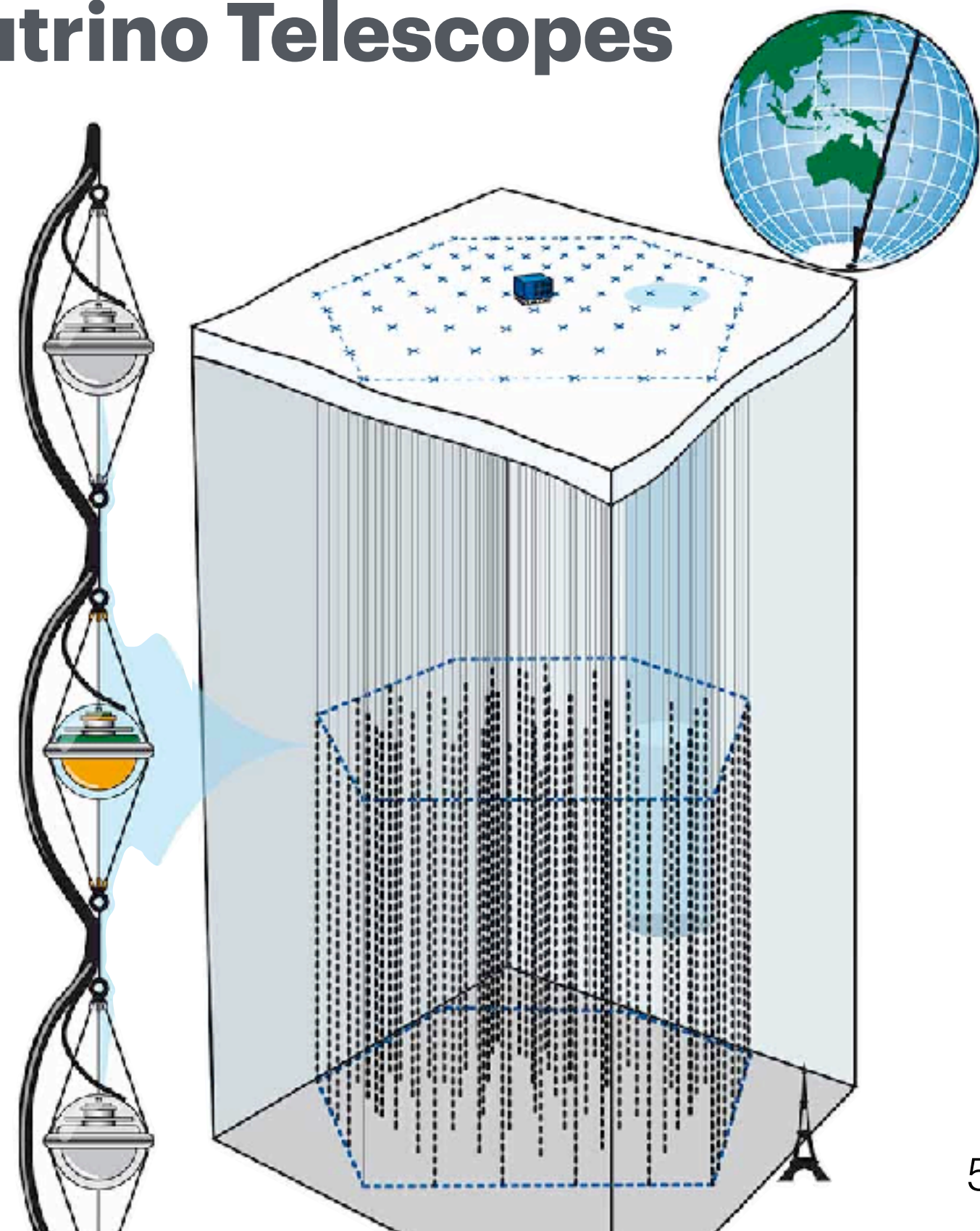


Reactors / Spallation Sources

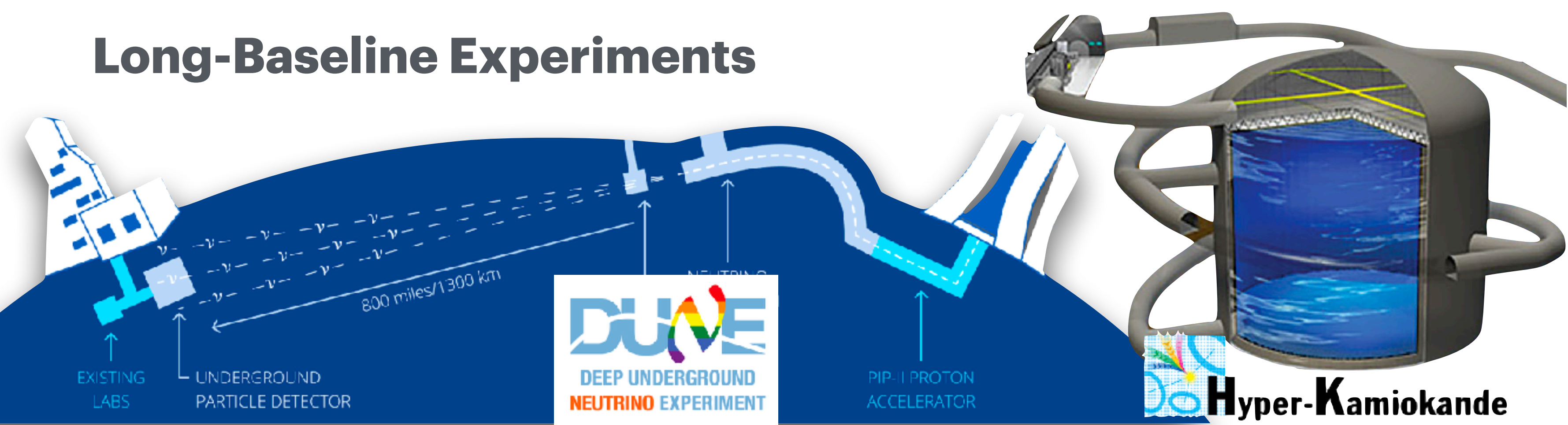
Adapted from P. Ochoa-Ricoux, NPN 2025



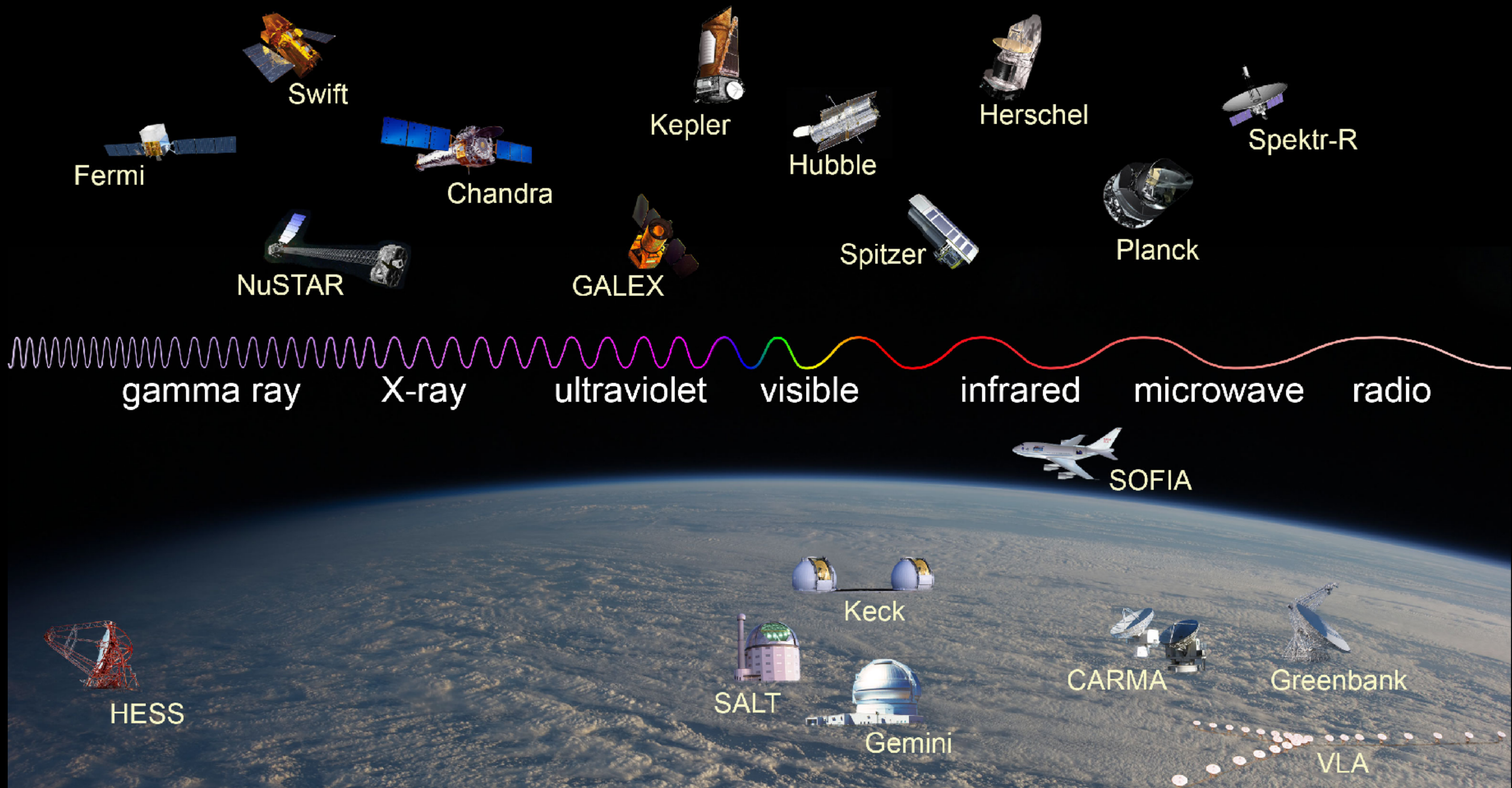
Neutrino Telescopes



Long-Baseline Experiments



Hyper-Kamiokande

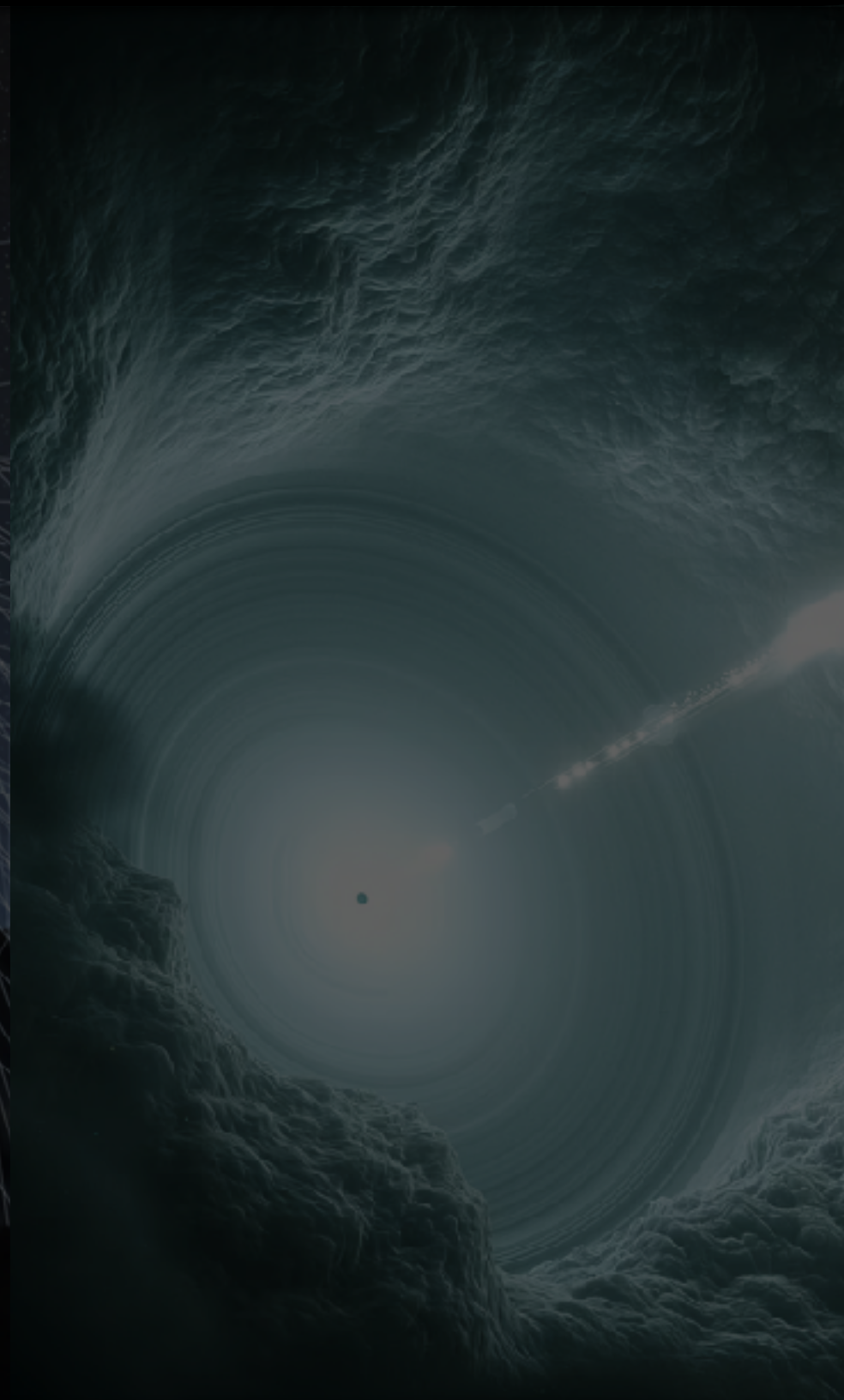
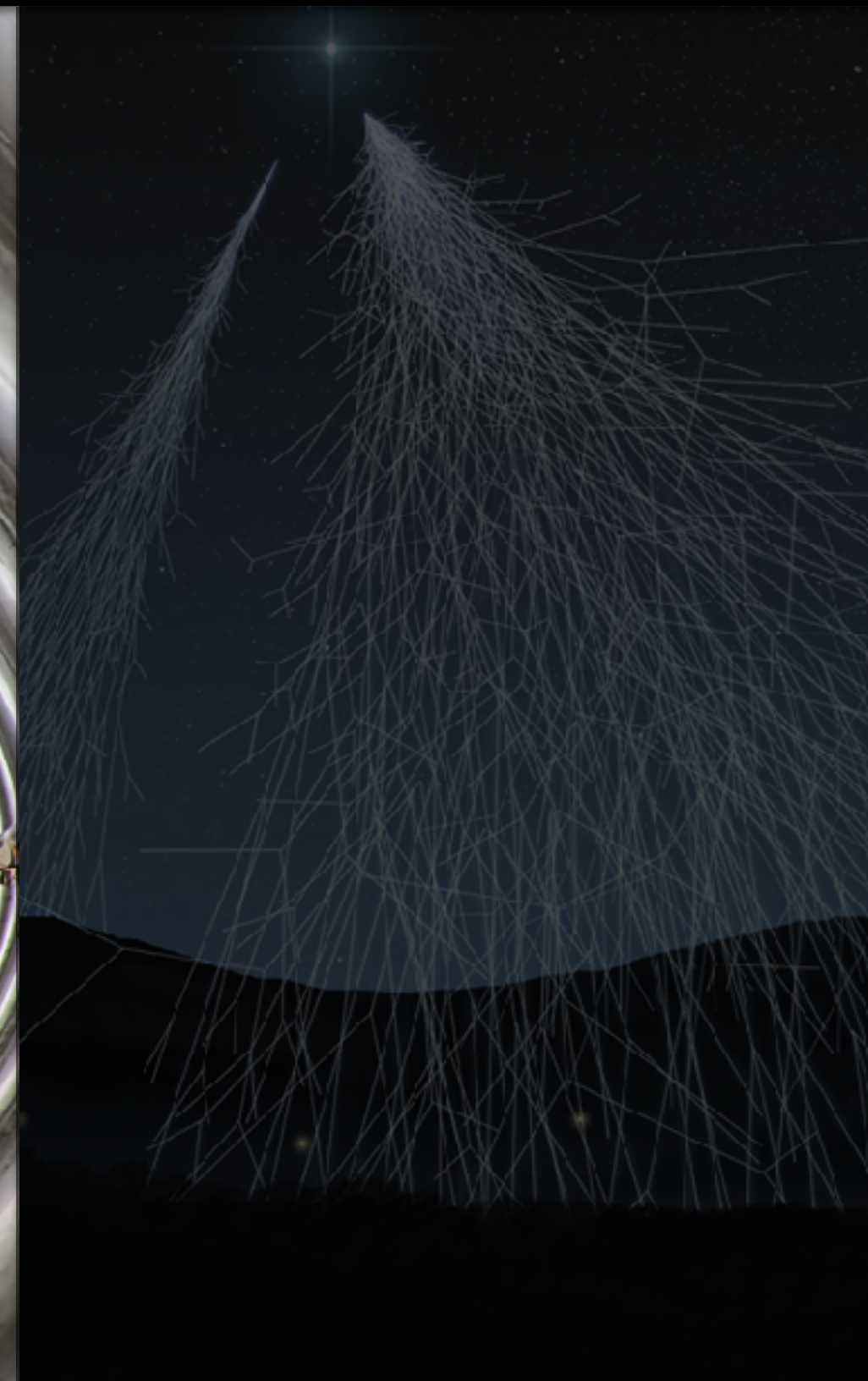
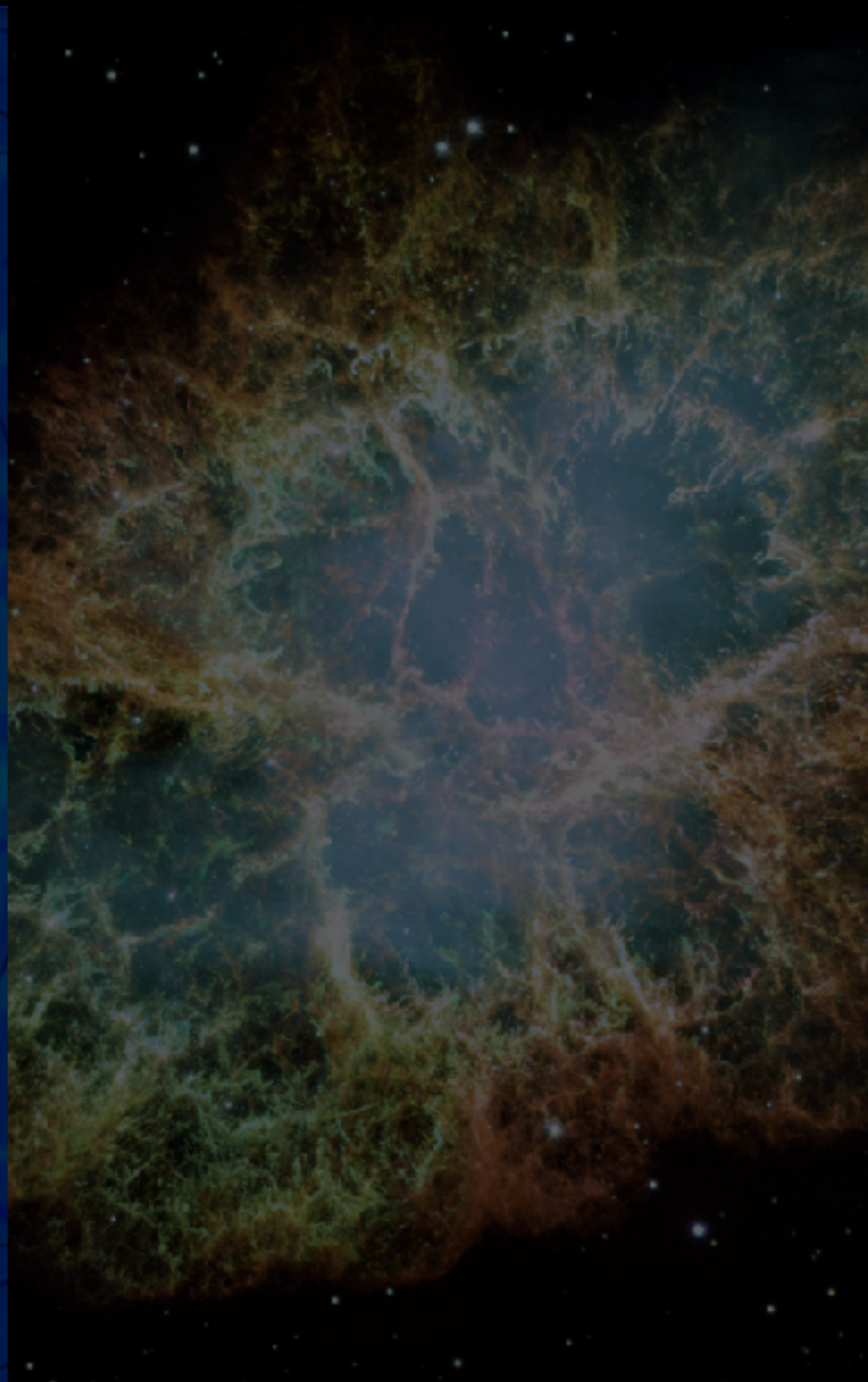
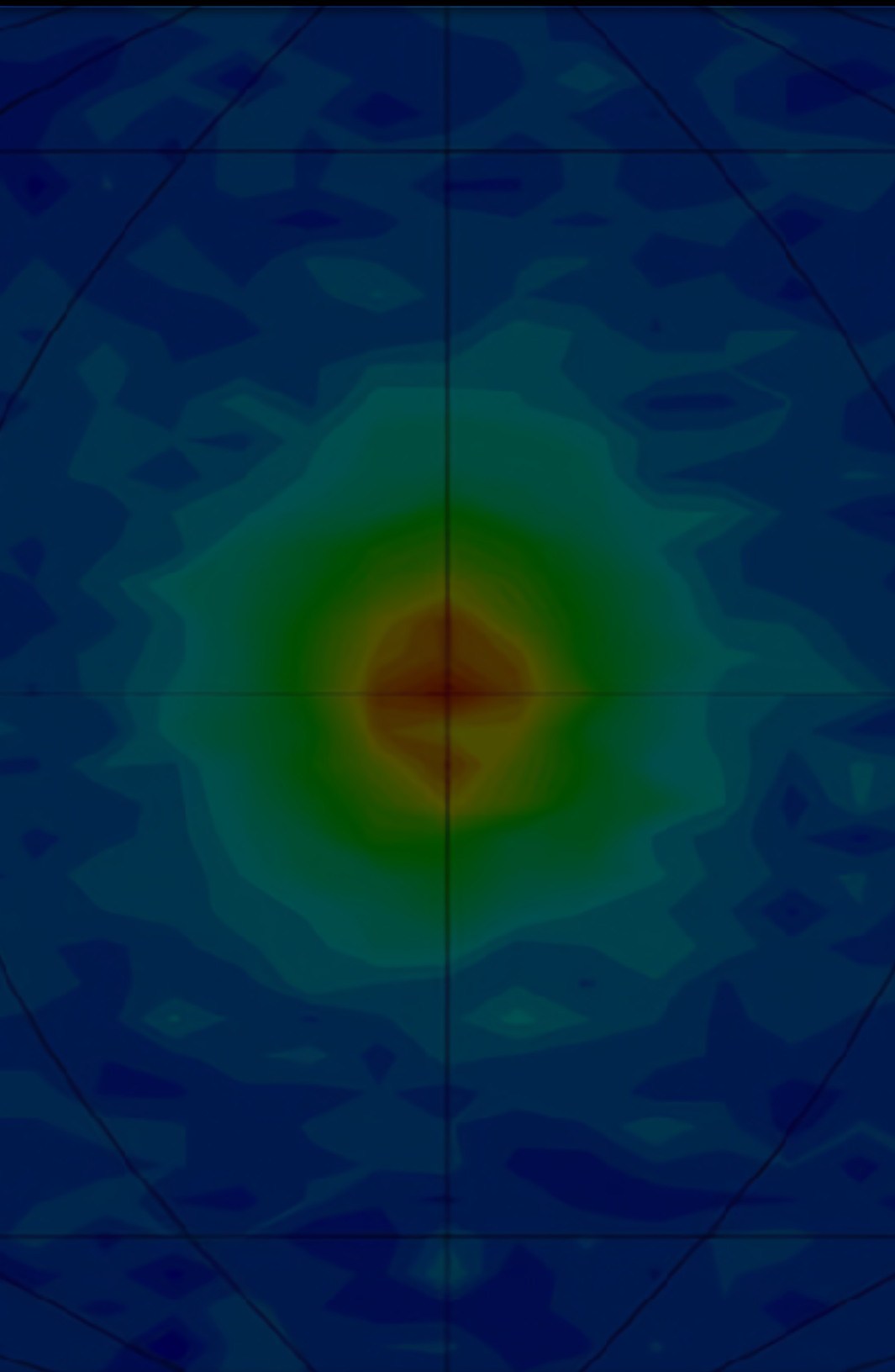


This talk:

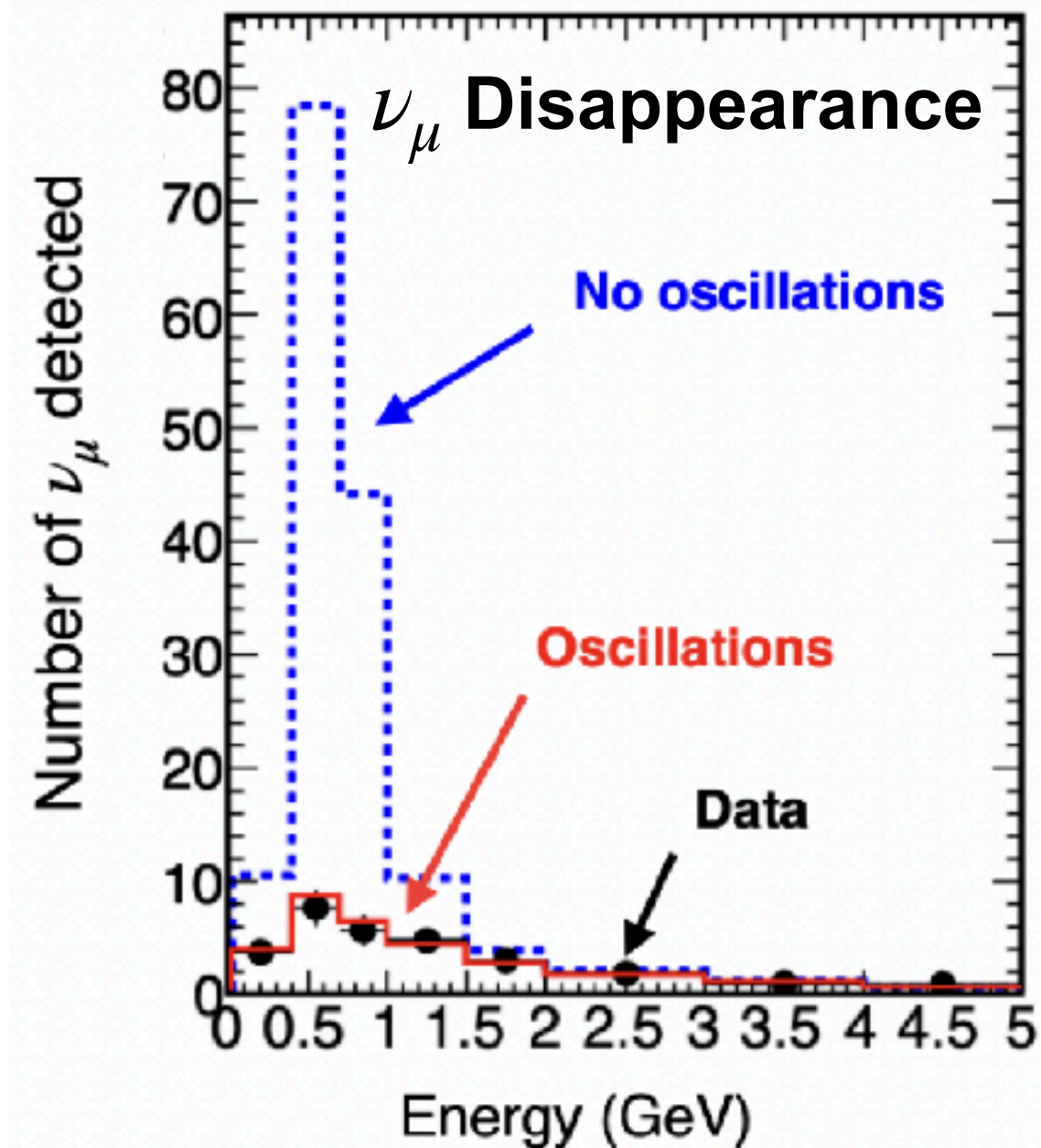
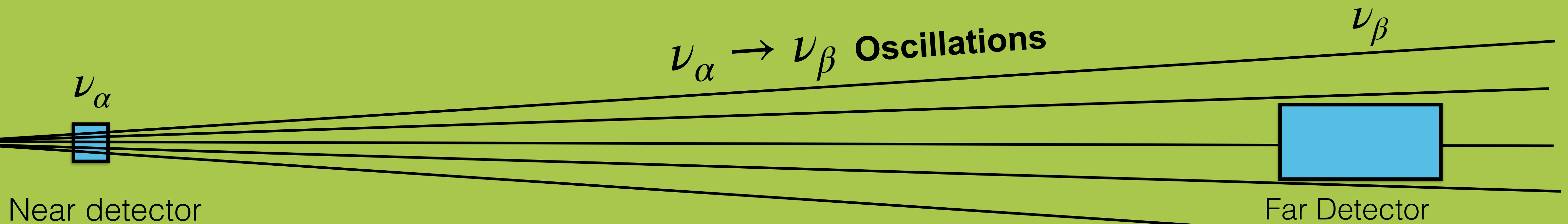
A theorist's perspective on **a few aspects of**
beyond the Standard Model physics with **neutrinos**

- 1) Sterile neutrinos at lab-based accelerators
- 2) Other weakly-interacting particles at accelerators
- 3) Ultra-long-distance neutrino propagation

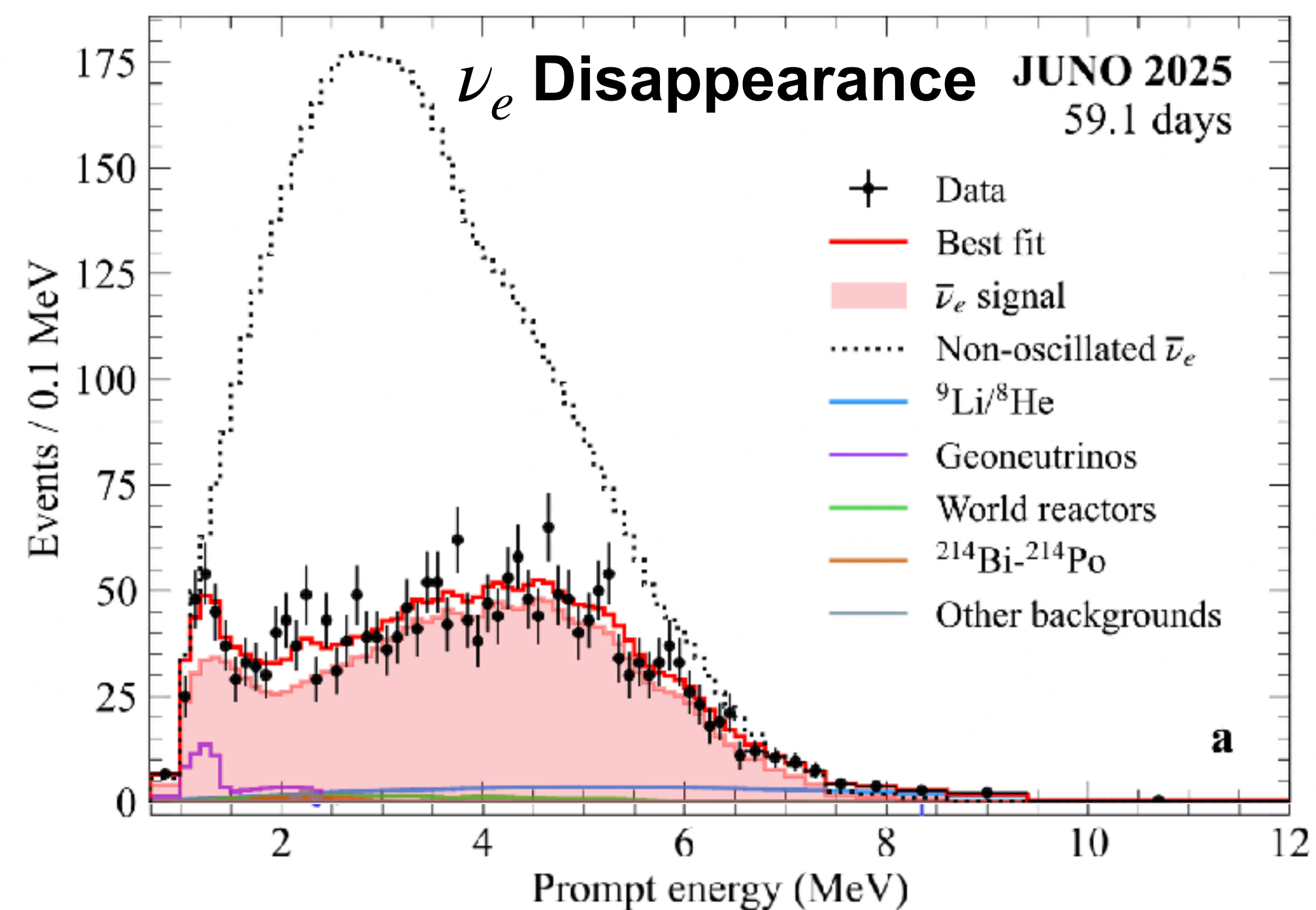
Rare Events at Accelerator Experiments



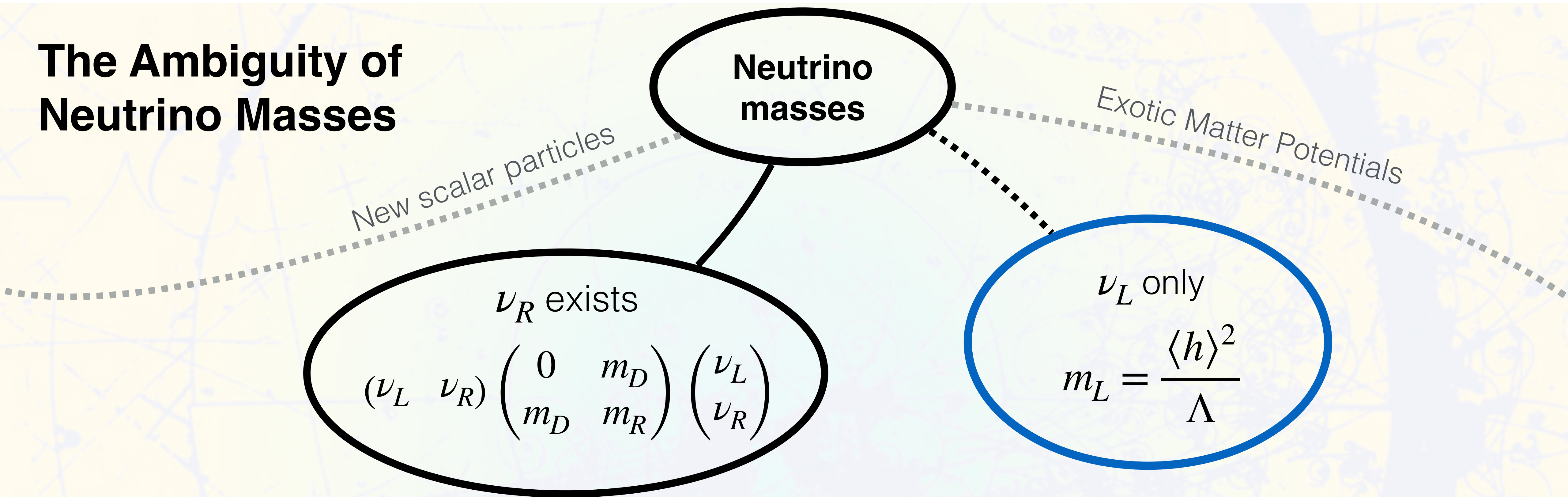
“Earth-Sized Quantum Flavor Interferometers”



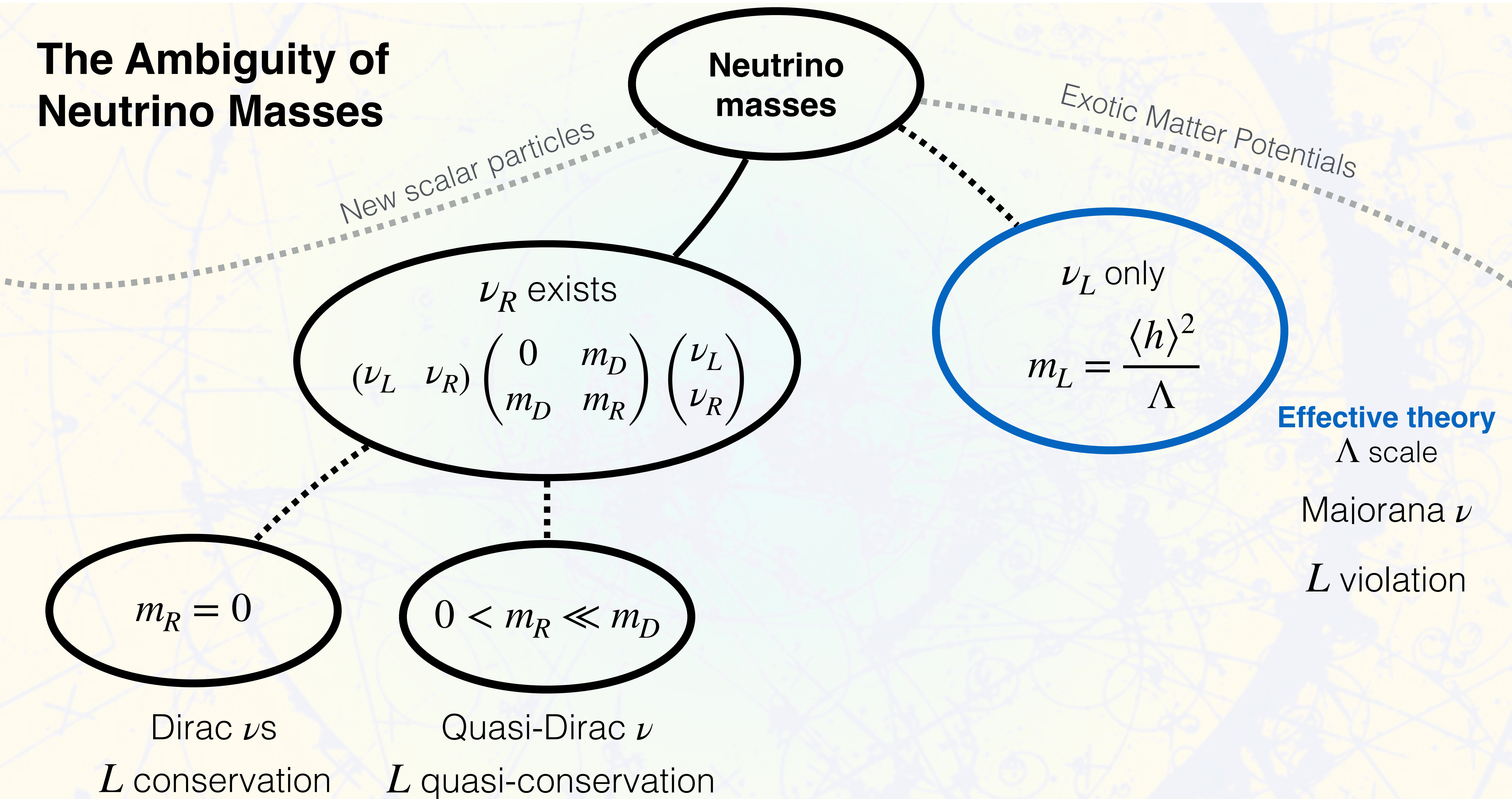
Adapted from T2K coll., [10.1103/PhysRevD.96.011102](https://arxiv.org/abs/10.1103/PhysRevD.96.011102)



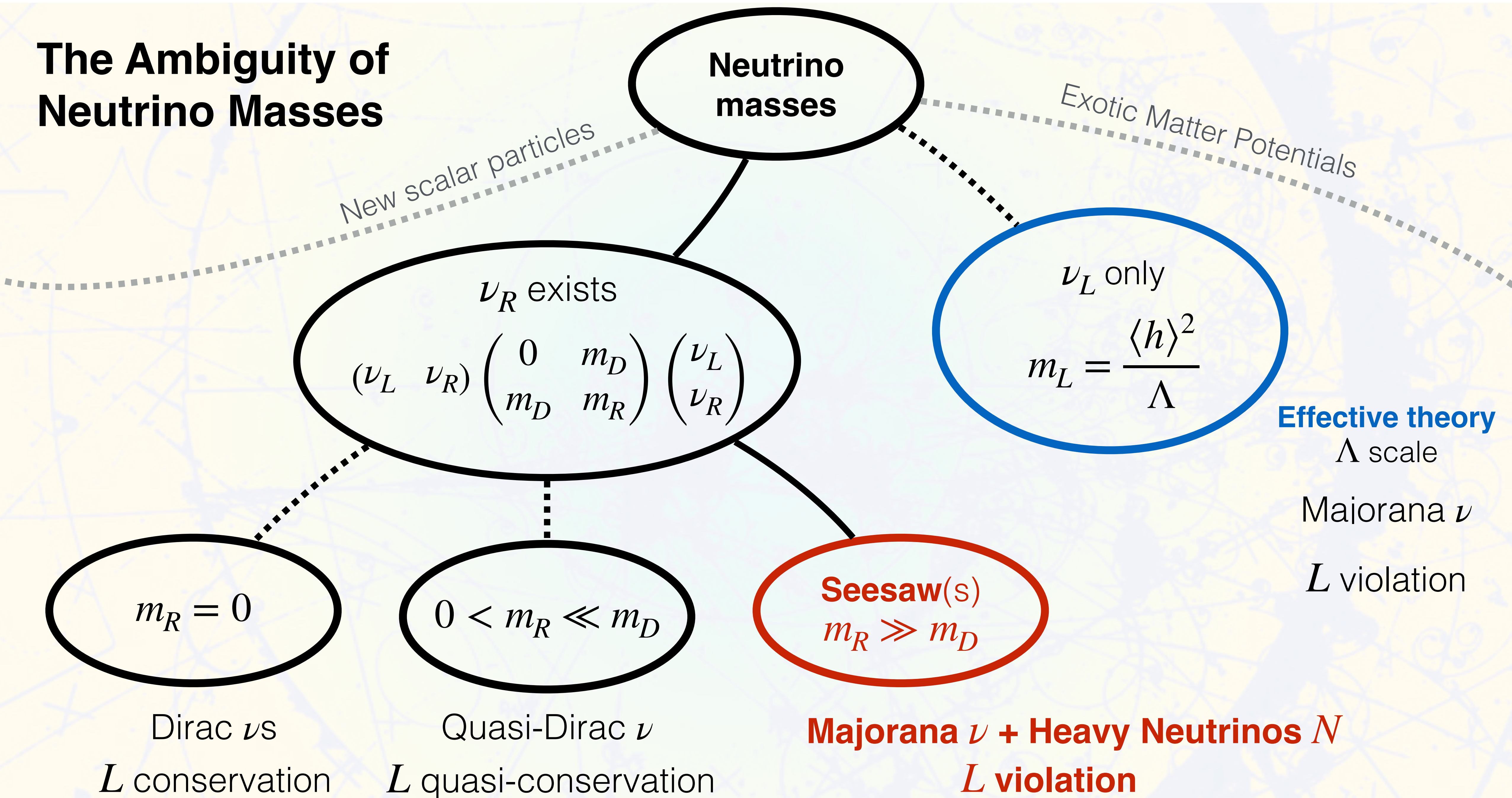
The Ambiguity of Neutrino Masses



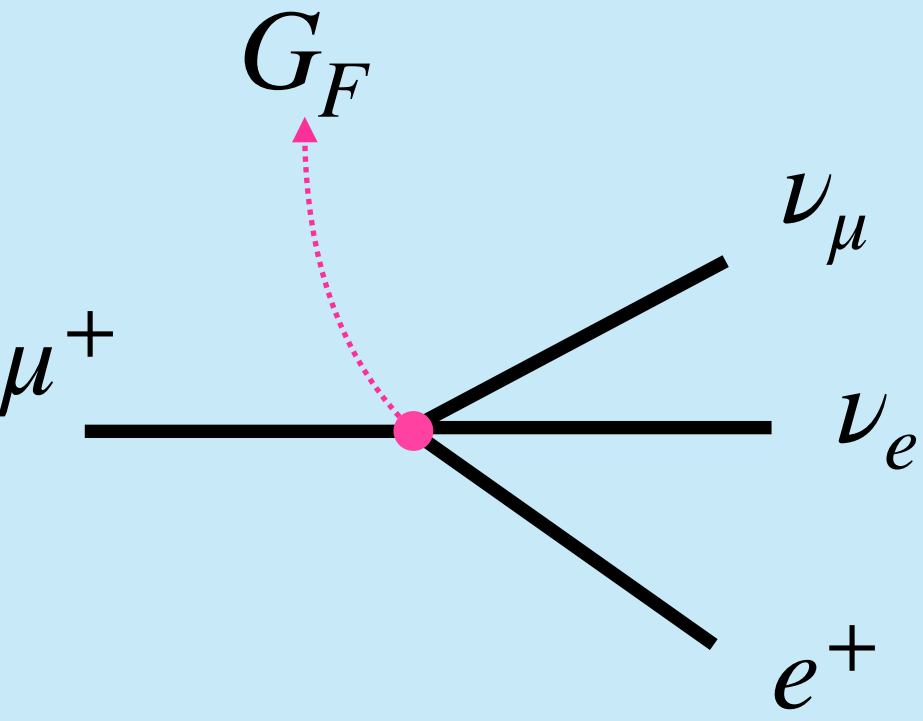
The Ambiguity of Neutrino Masses



The Ambiguity of Neutrino Masses

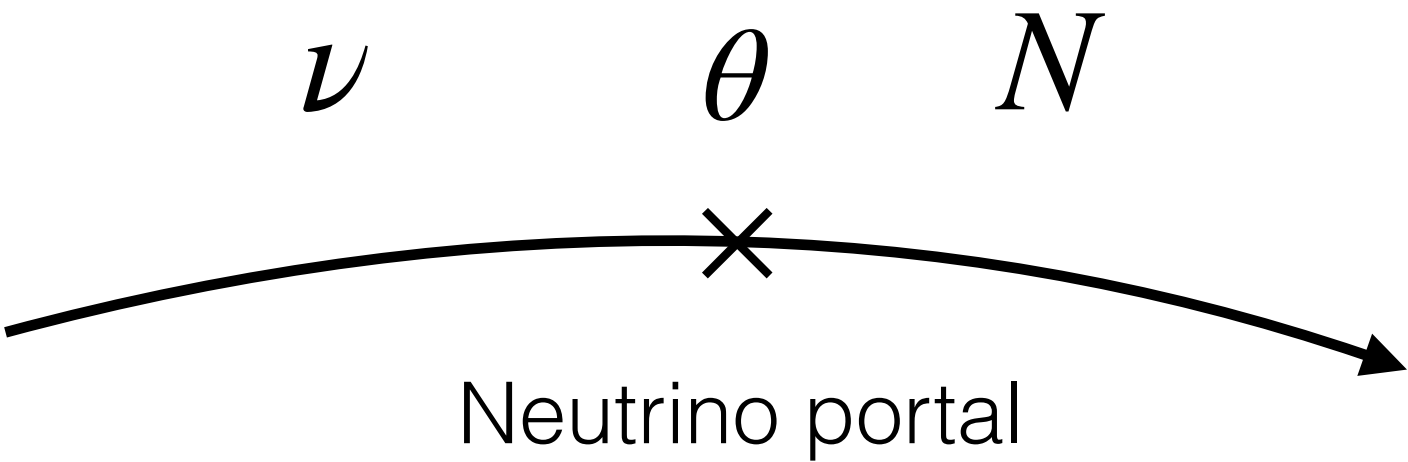


**STANDARD
MODEL**



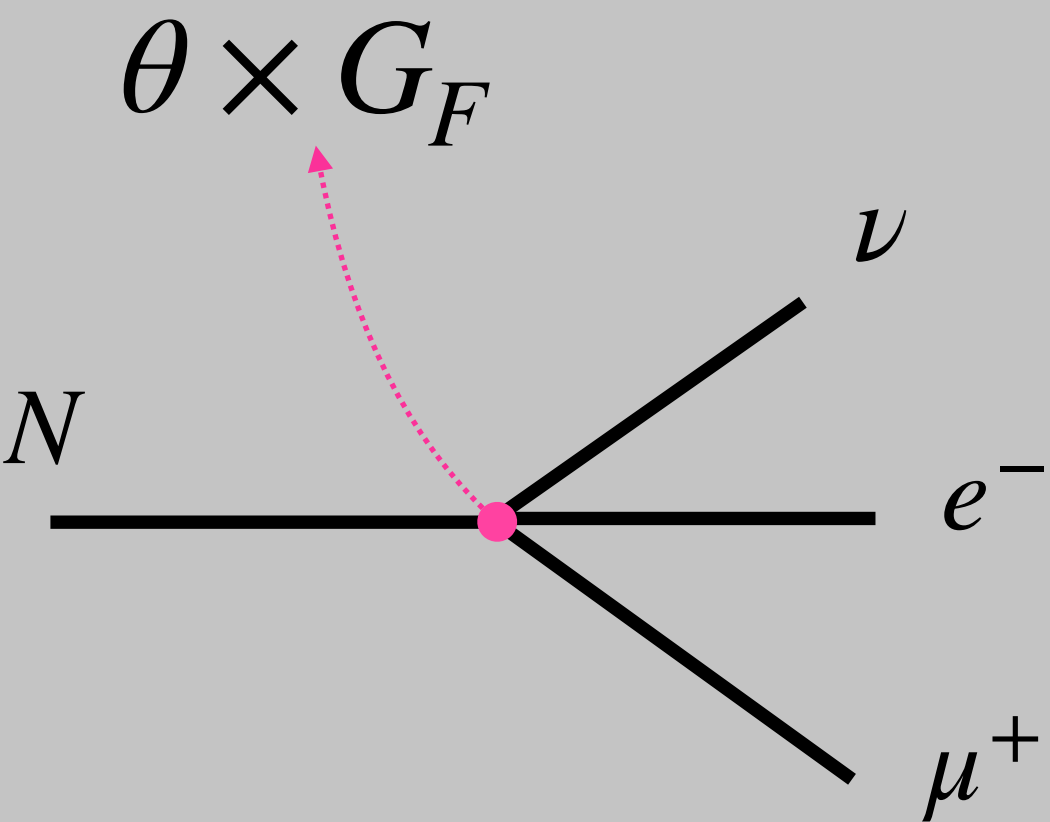
Weak interaction

$$G_F \sim \frac{1}{m_W^2}$$



Heavy Neutrinos

HIDDEN SECTOR



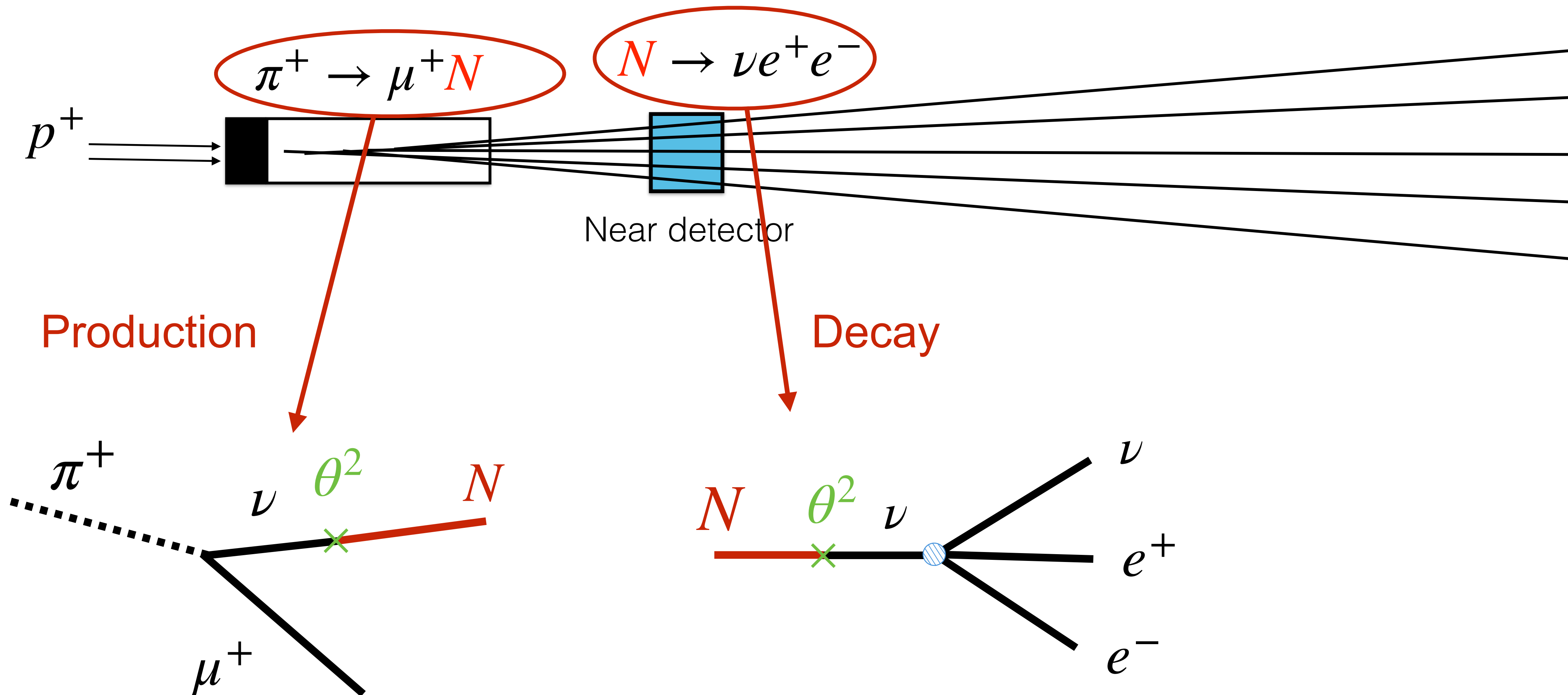
Weaker-than-Weak interactions
 $(\theta \ll 1)$

Long-lived particle

$$\Gamma \sim \theta^2 G_F^2 M_N^5$$

Accelerator Neutrino Experiments

Near detectors / short-baselines

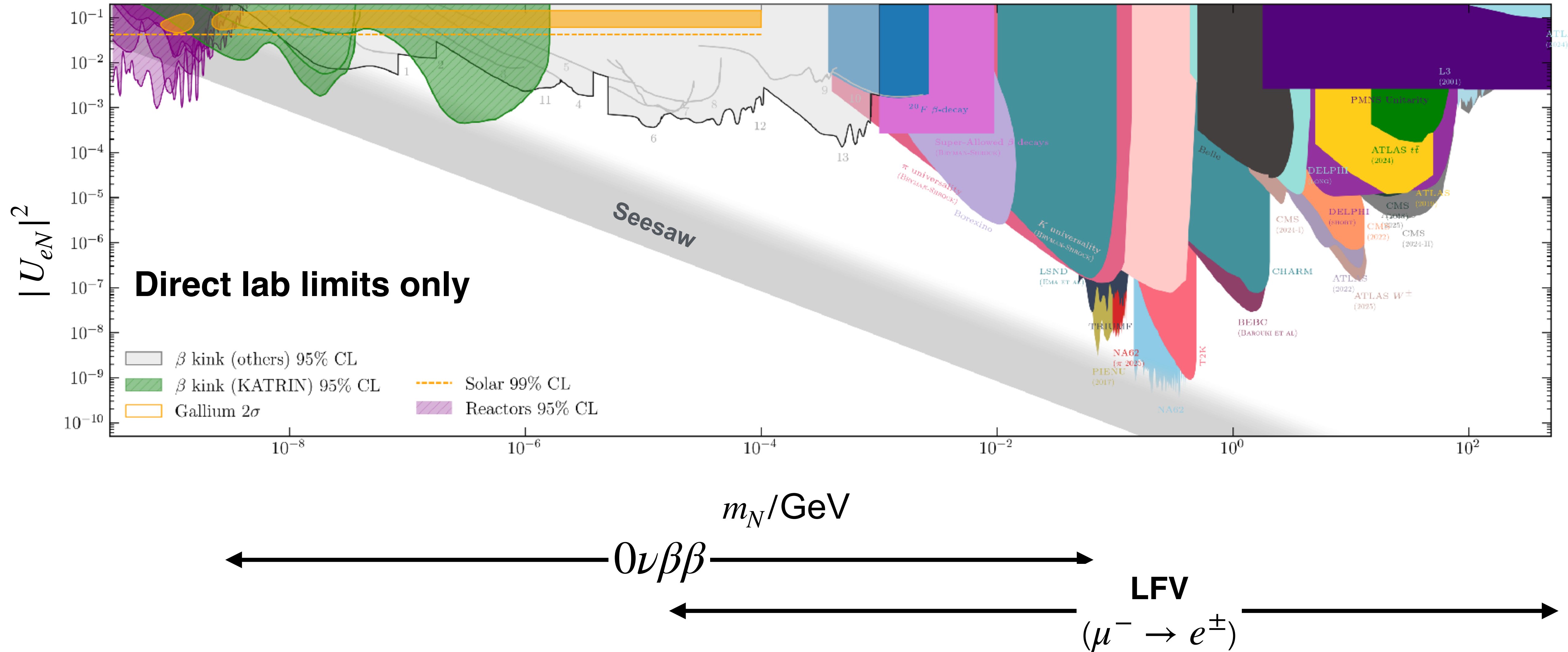


3+1 oscillations

Cosmo & β -decay

Decays in flight &
Rare meson decays

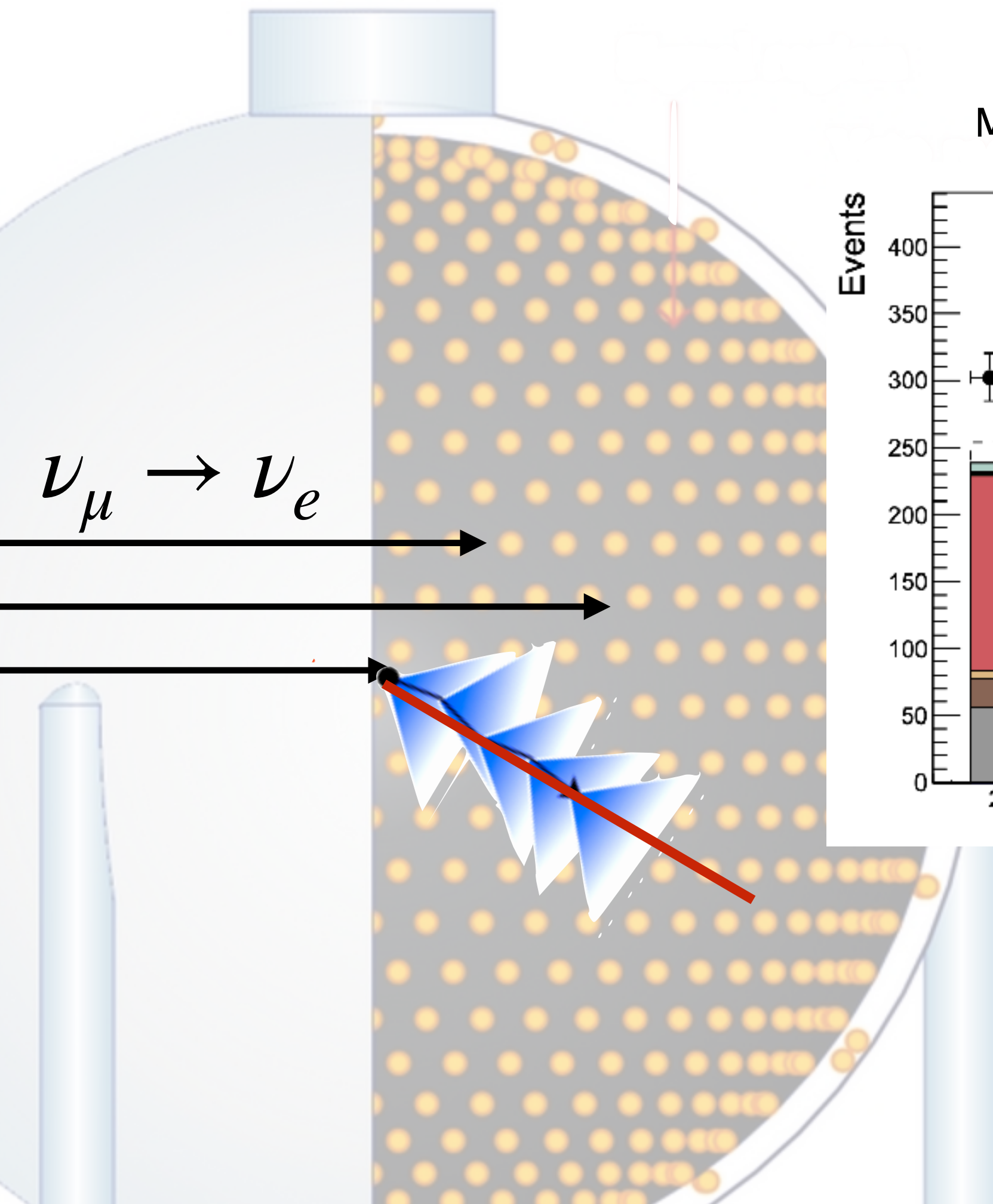
Colliders



The Short-Baseline Puzzle: appearance signals?

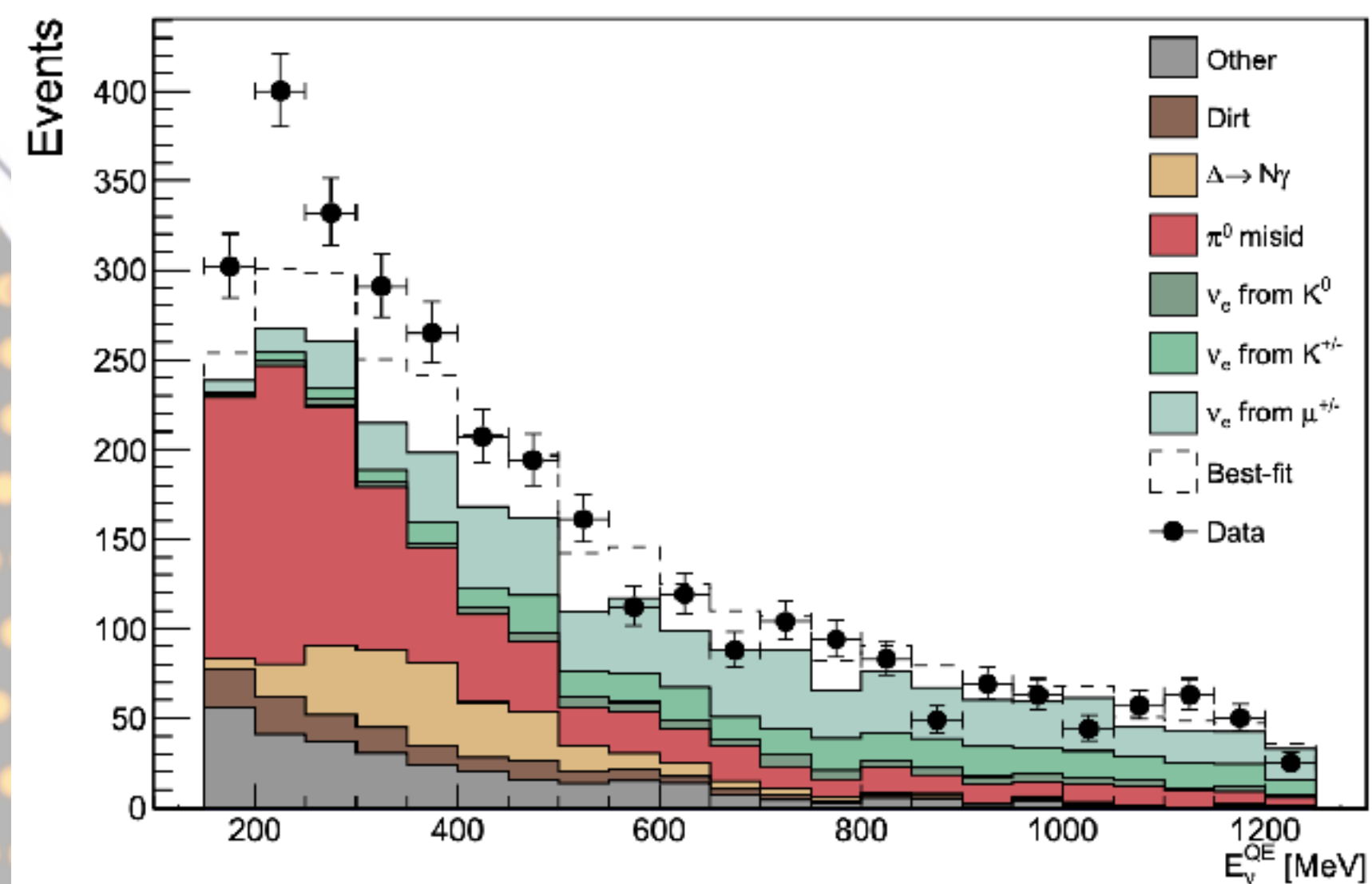
LSND

MiniBooNE detector



MiniBooNE

MiniBooNE coll., PRD 103, 052002 (2021)



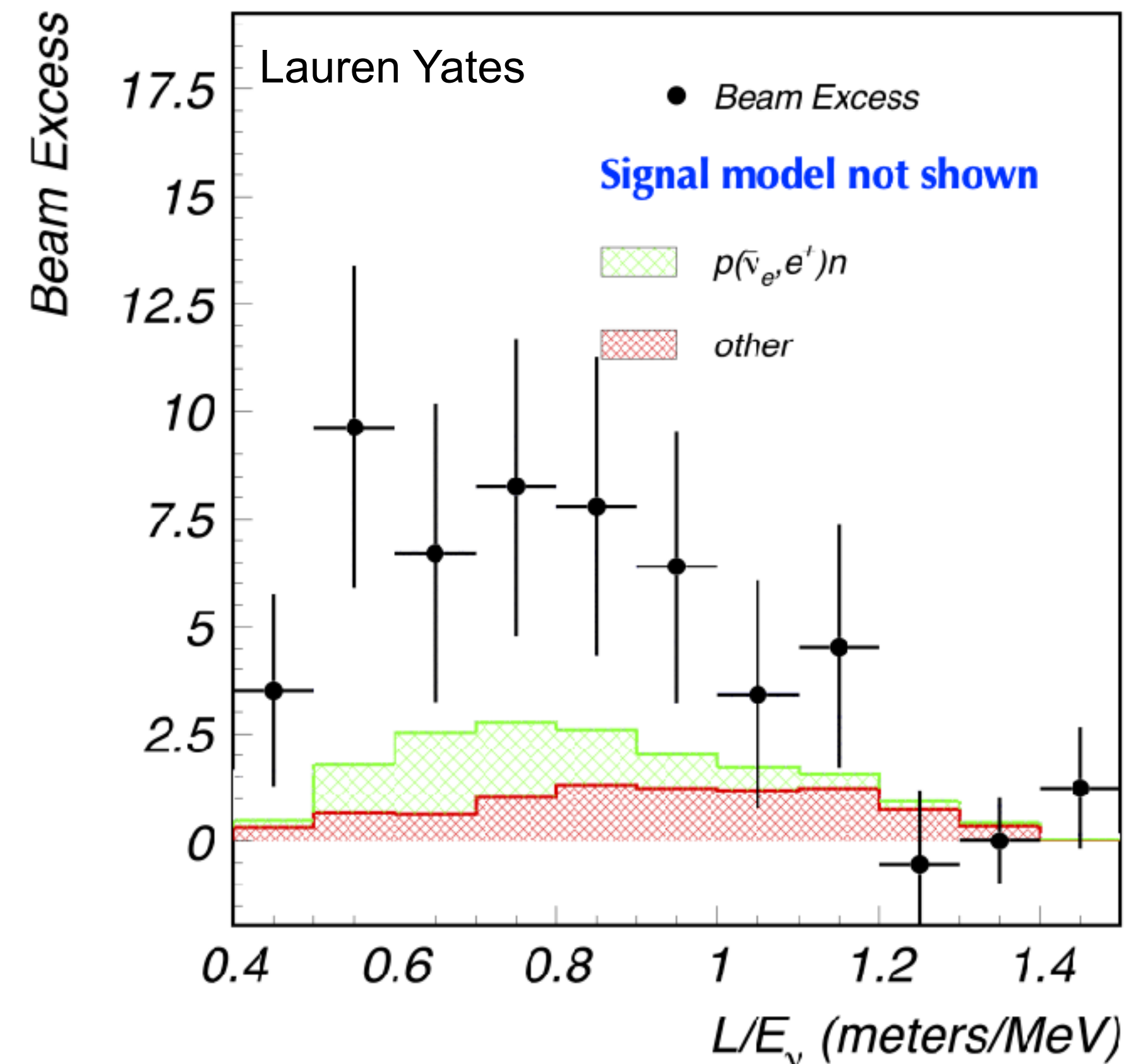
Reconstructed E_{ν}^{QE} / MeV

Excess:

$638 \pm 52(\text{stat.}) \pm 122.2(\text{sys.})$

4.8σ significance

LSND coll. PRD 64 (2001) 112007

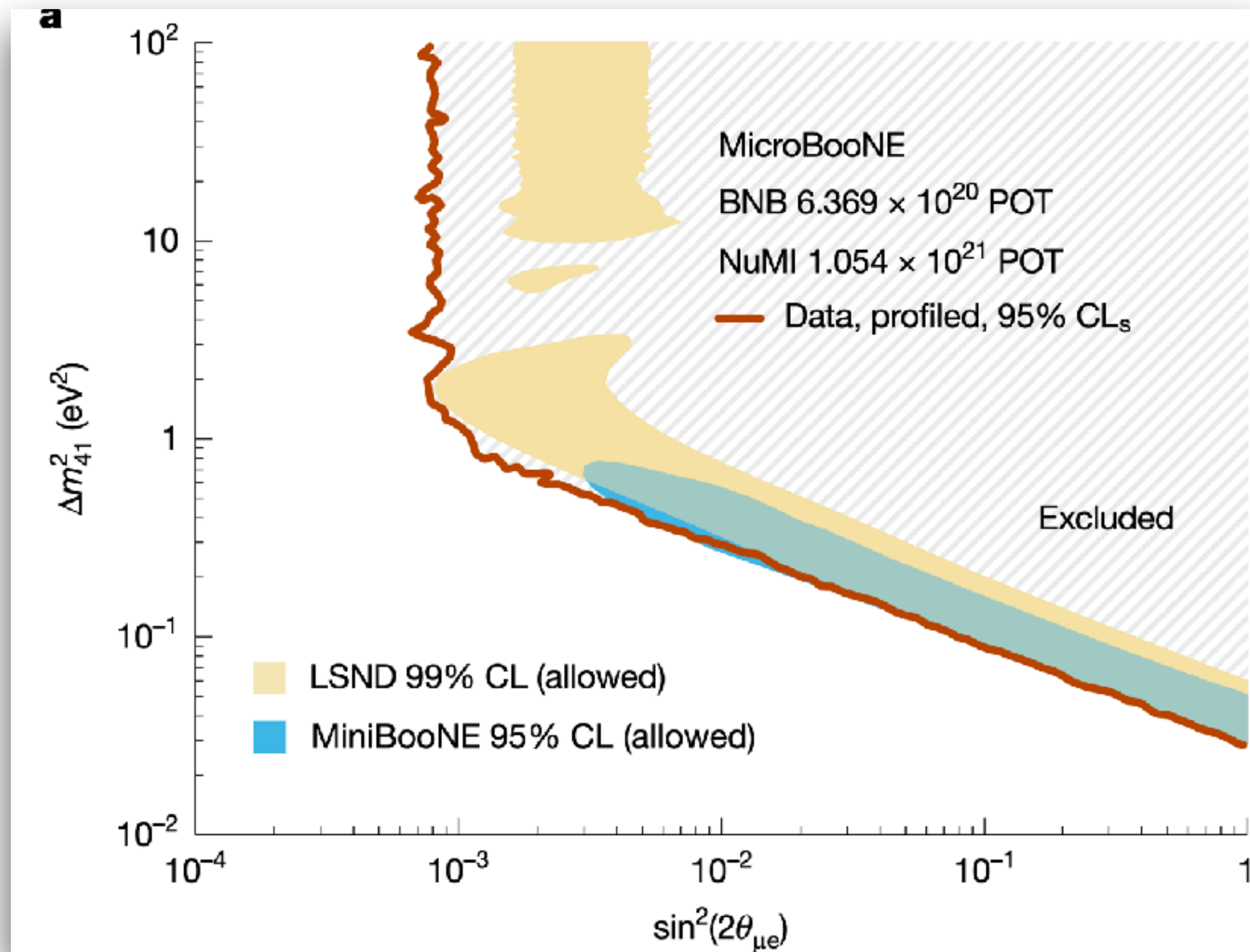


Excess: $87.9 \pm 22.4 \pm 6$ events

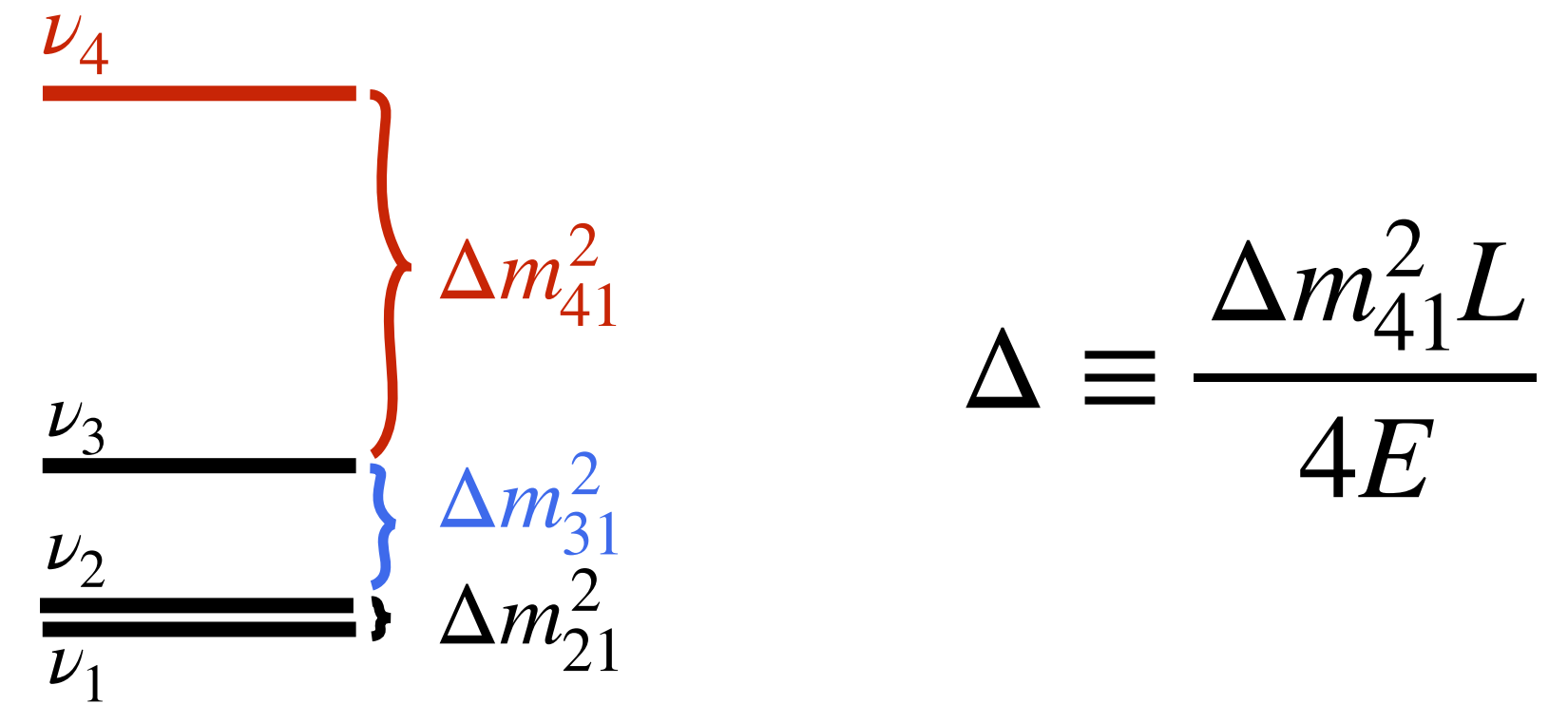
3.8σ significance

Light Sterile Neutrino Oscillations

Latest results from MicroBooNE experiment



Article | [Open access](#) | Published: 03 December 2025
Search for light sterile neutrinos with two neutrino beams at MicroBooNE
[The MicroBooNE Collaboration](#)
[Nature](#) 648, 64–69 (2025) | [Cite this article](#)



Appearance

$$P_{\nu_{\mu} \rightarrow \nu_e} = \sin^2 2\theta_{e\mu} \sin^2 \Delta = 4 |U_{e4}|^2 |U_{\mu4}|^2 \sin^2 \Delta$$

Fresh off the Press

First time the LSND + MiniBooNE regions are fully covered using a single detector.
 (1 detector, 2 beams!)

No resolution to the anomalies, but minimal model is disfavored by global oscillation picture.

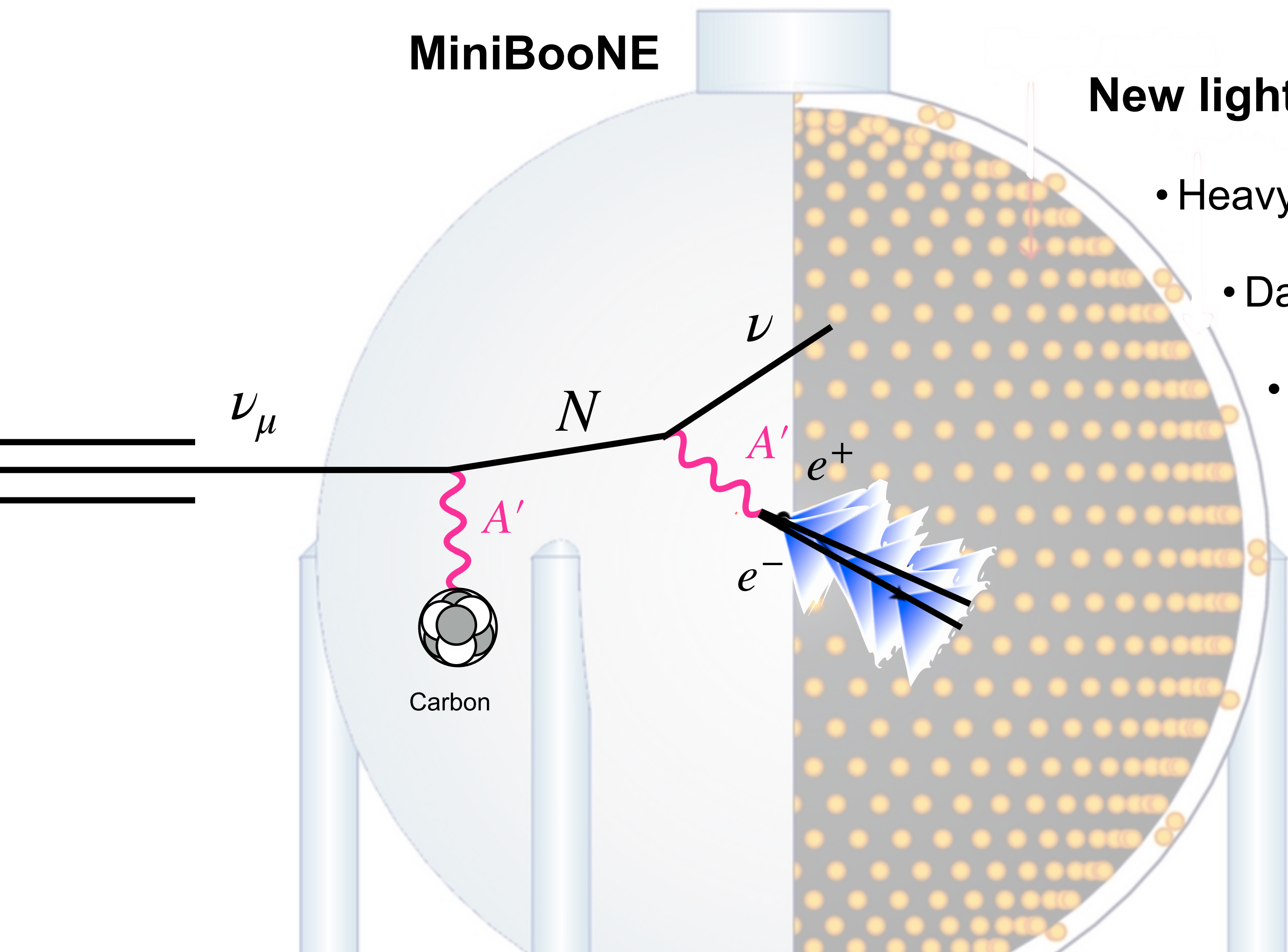
Many questions remain open.

Dark Neutrinos

A multi-portal exploration of dark sectors

E. Bertuzzo, S. Jana, P. A. N. Machado, R. Z. Funchal, [PRL 121, 241801 \(2018\)](#)
P. Ballett, M. Ross-Lonergan, S. Pascoli, [PRD 99, 071701 \(2019\)](#)

C. Argüelles, MH, Y. Tsai, [PRL 123, 261801 \(2019\)](#)
P. Ballett, MH, S. Pascoli, [PRD 101, 115025 \(2020\)](#)



New light particles produced in neutrino interactions:

- Heavy Neutrinos?
- Dark Photons?
- Dark Higgses?



Dark Neutrinos @ MicroBooNE

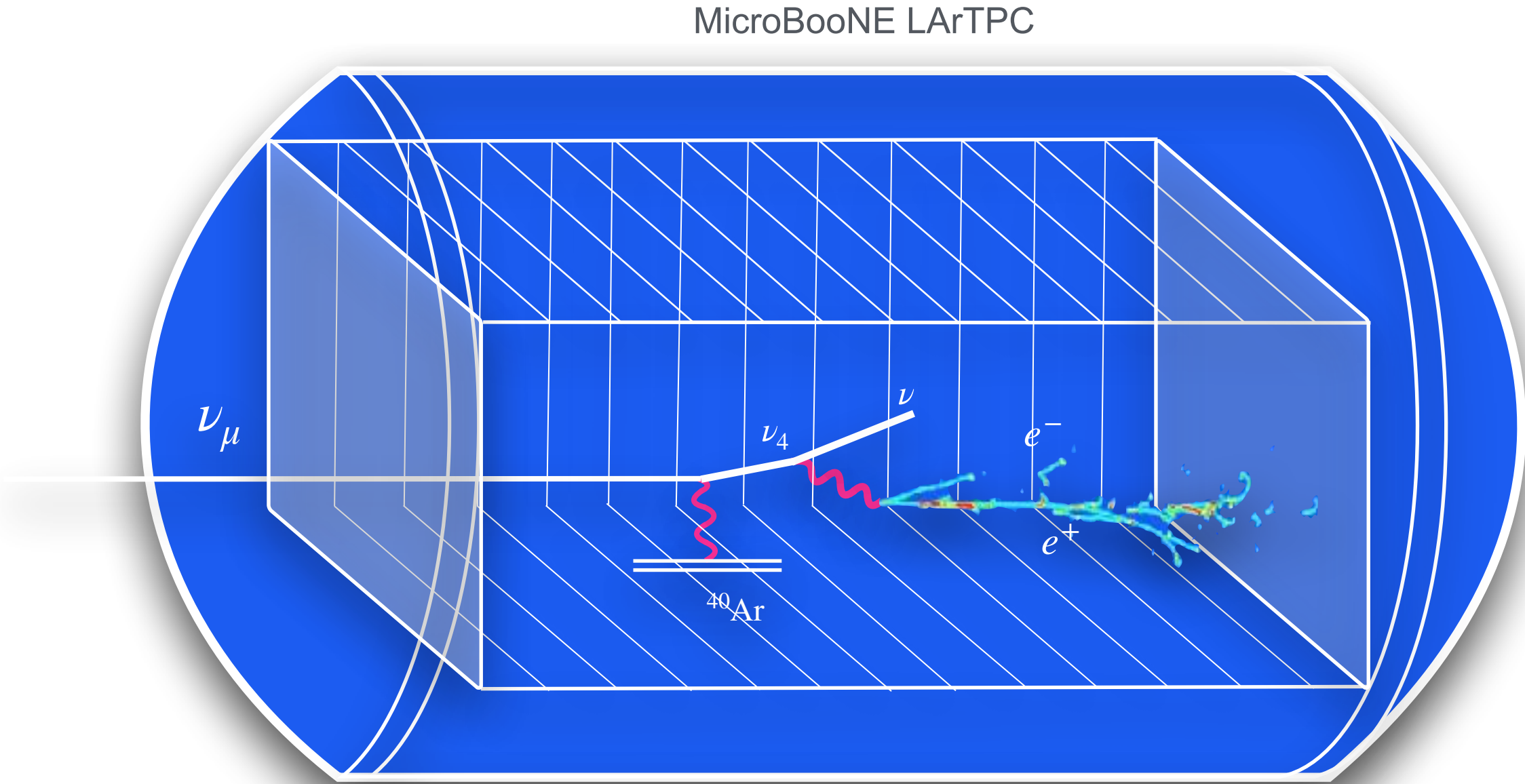
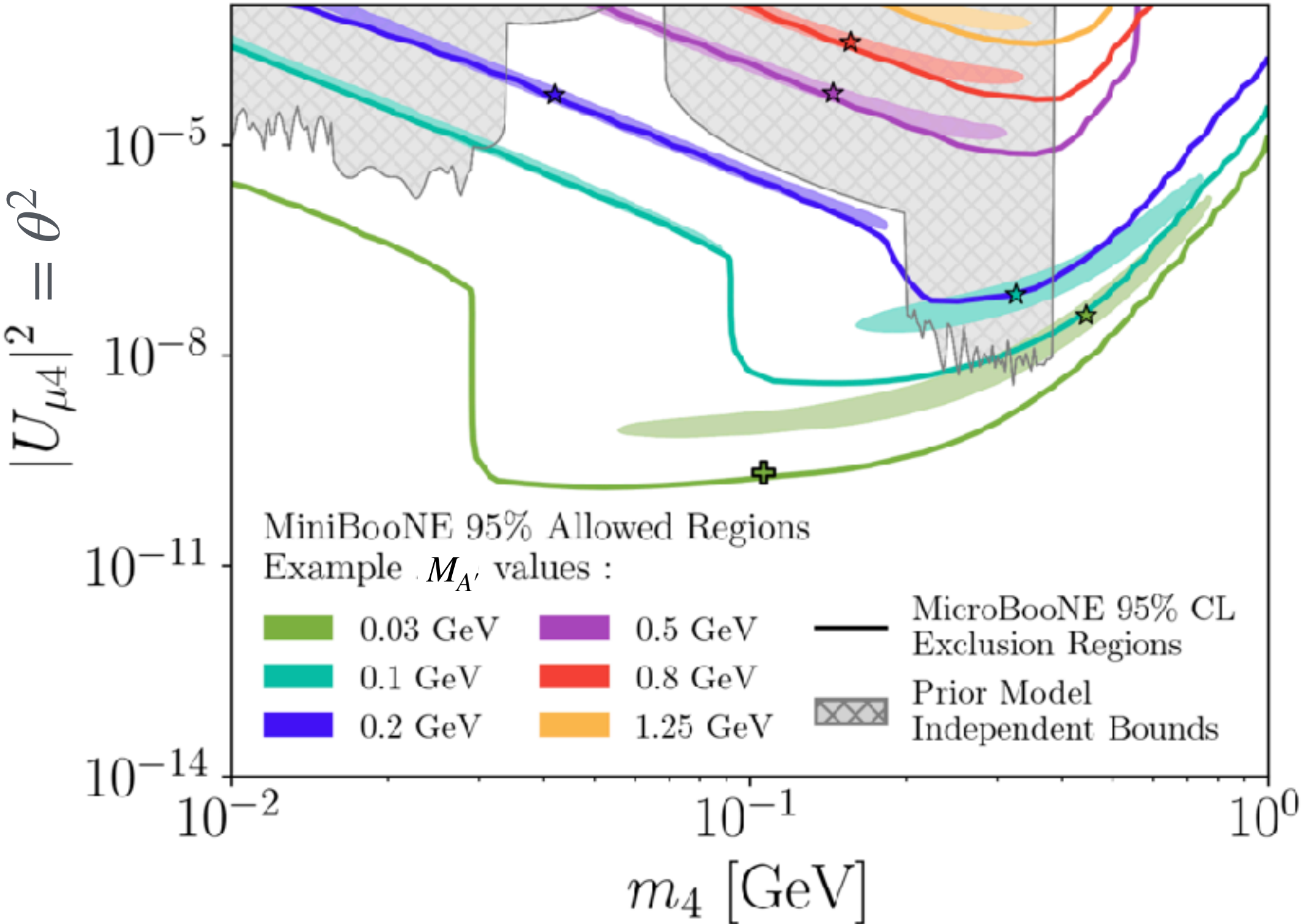
First (modern) exp limits on neutrino upscattering

MicroBooNE coll. +
A. Abdullahi, MH, D. Massaro, S. Pascoli, J. Zink.
arxiv.org/abs/2502.10900

First dedicated experimental search for neutrino-induced e^+e^- pairs:

MicroBooNE excludes minimal dark neutrino explanations to the MiniBooNE anomaly

E. Bertuzzo, S. Jana, P. A. N. Machado, R. Z. Funchal, [PRL 121, 241801 \(2018\)](#)
P. Ballett, M. Ross-Lonergan, S. Pascoli, [PRD 99, 071701 \(2019\)](#)



More Opportunities at the Booster Neutrino Beam

SBND and ICARUS will have even more statistics to search for light particles:

SBND: $\sim 7,000$ neutrino interactions /day!

In addition, several opportunities for antineutrino or beam-dump mode running modes being discussed.

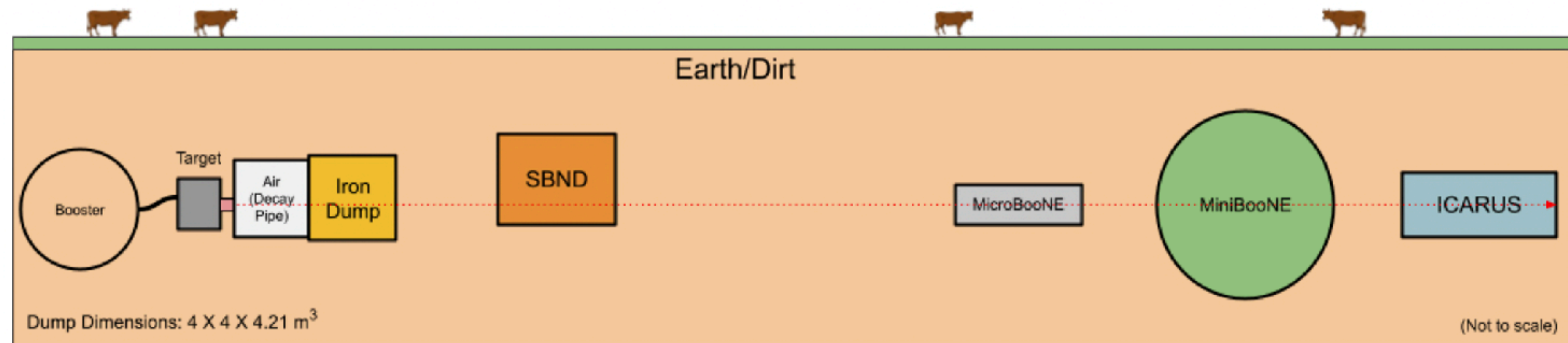
Beam dump mode would enable significantly more light particle production and potentially meson decay-at-rest measurement at ~ 100 t LArTPC

Input to European Strategy for Particle Physics 2026 Update

Enhancing New Physics Searches with a Future Beam Dump Configuration at SBND^a

B. Dutta,^{1,†} D. Goswami,^{1,‡} A. Karthikeyan,^{1,§}
V. Pandey,^{2,¶} Z. Tabrizi,^{3,4,**} and R. G. Van de Water^{5,††}

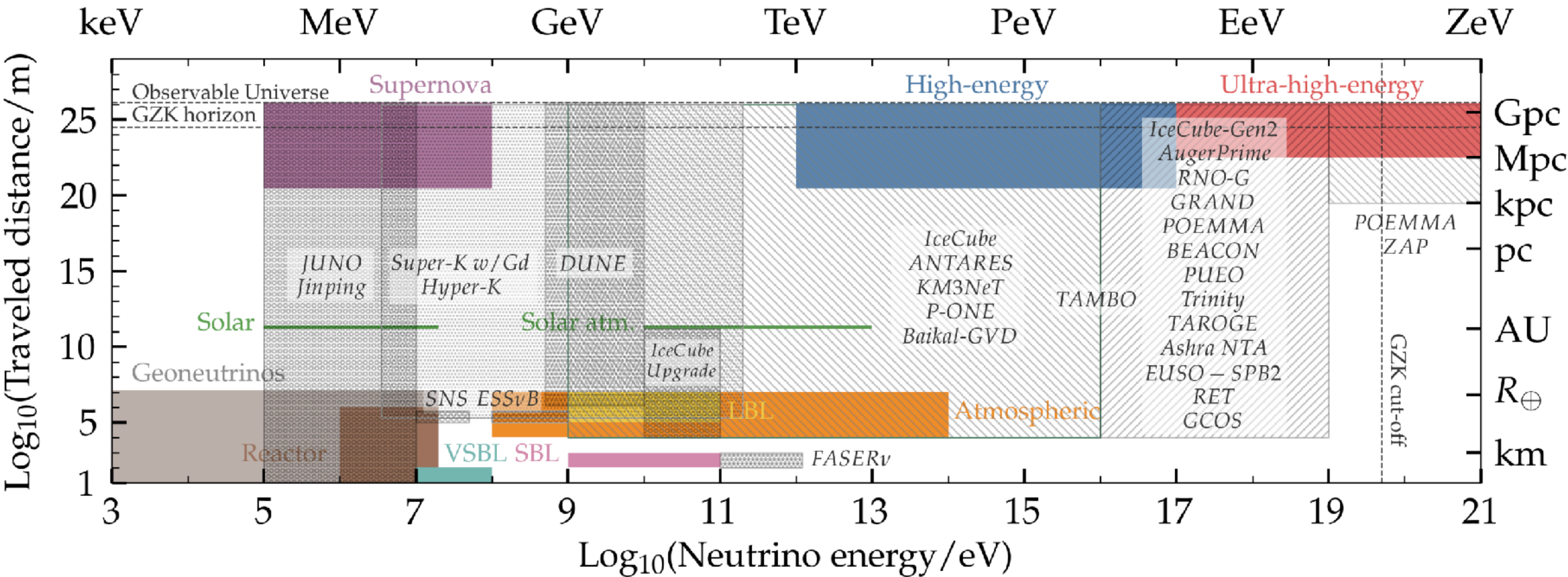
B. Dutta, D. Goswami, A. Karthikeyan, K. J. Kelly, [JHEP 05 \(2025\) 240](#)



Ultra-long-distance neutrino propagation



Ultra-long baseline propagation



$$e^{-i(E_1-E_2)t} \longrightarrow e^{-i\frac{\Delta m^2_t}{2E}} \times e^{-it\epsilon_{\text{new physics?}}}$$



Neutrino Spin Flavor Precession

Transition magnetic moment couples to background B-field to precess the neutrino spin and leads to:

$$\nu_e^L \rightarrow \bar{\nu}_\mu^R \text{ (for Majorana } \nu)$$

For Majorana, have flavor off-diagonal moments:

$$\mathcal{L} = \frac{\mu}{2} \bar{\nu}_e \sigma_{\mu\nu} F^{\mu\nu} \nu_\mu$$

See for example:

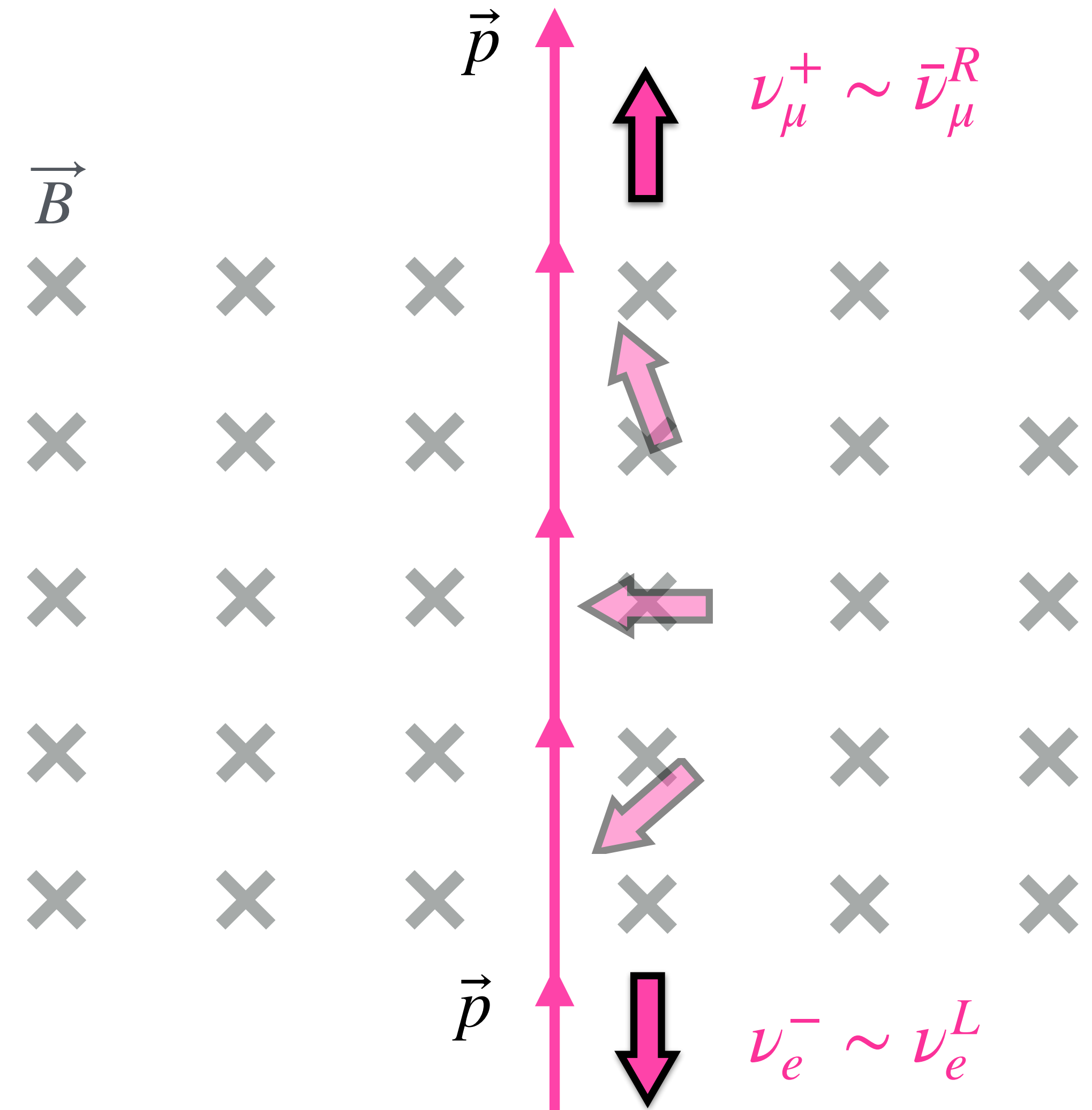
J. Kopp, T. Opferkuch, E. Wang, [JCAP 03 \(2024\) 043](#)

V. Brdar, T. Cheng, H-J. Kuan, Y-Y. Li, [JCAP 07 \(2024\) 026](#)

Resonant spin-flavor rotation:

C.S. Lim, W. J. Marciano (1988)

E. Akhmedov (1988)



Propagation of Majorana Neutrinos through an Ultra-Light Dark Matter Background

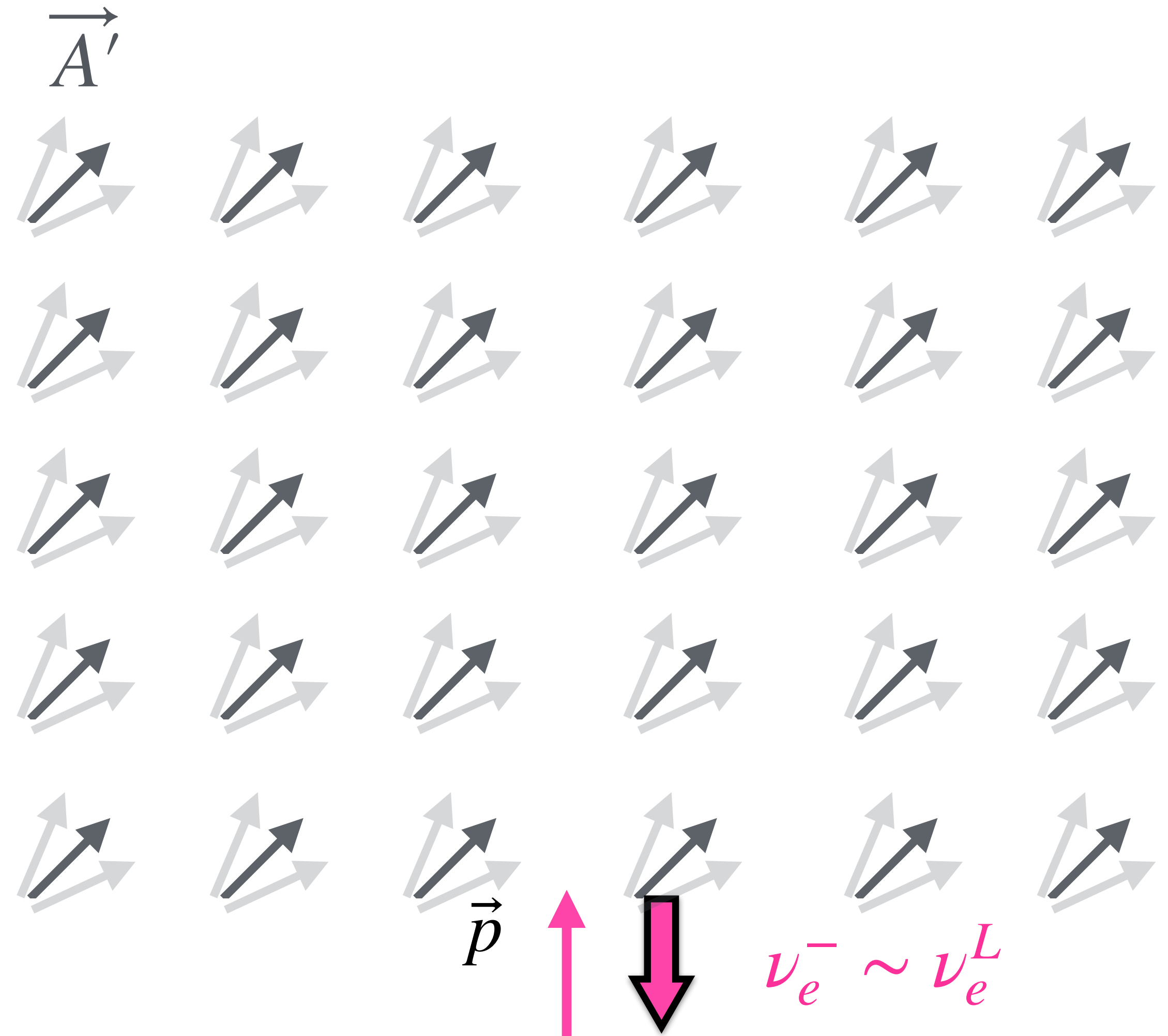
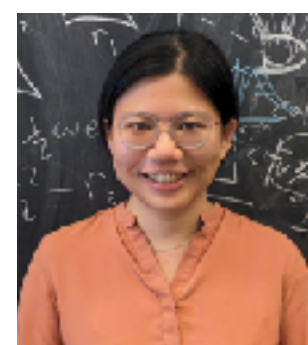
A Dark Vector Dark Matter background can couple to an axial-vector Majorana neutrino current:

$$\mathcal{L} \supset e' A'_\mu (\bar{\nu} \gamma^\mu \gamma^5 \nu)$$

Classical field background with $m_{A'} \ll 1$ eV.

$$\vec{A}'(t) \simeq \frac{\sqrt{2\rho_{\text{DM}}/3}}{m_{A'}} \text{Re} \left[e^{im_{A'}t} \sum_{i=1}^3 \alpha_i e^{i\phi_i} \hat{n}_i \right]$$

In progress and in collaboration with:
A. Berlin, R. Capdevilla, T. Cheng, P. Machado (FNAL)



Propagation of Majorana Neutrinos through an Ultra-Light Dark Matter Background

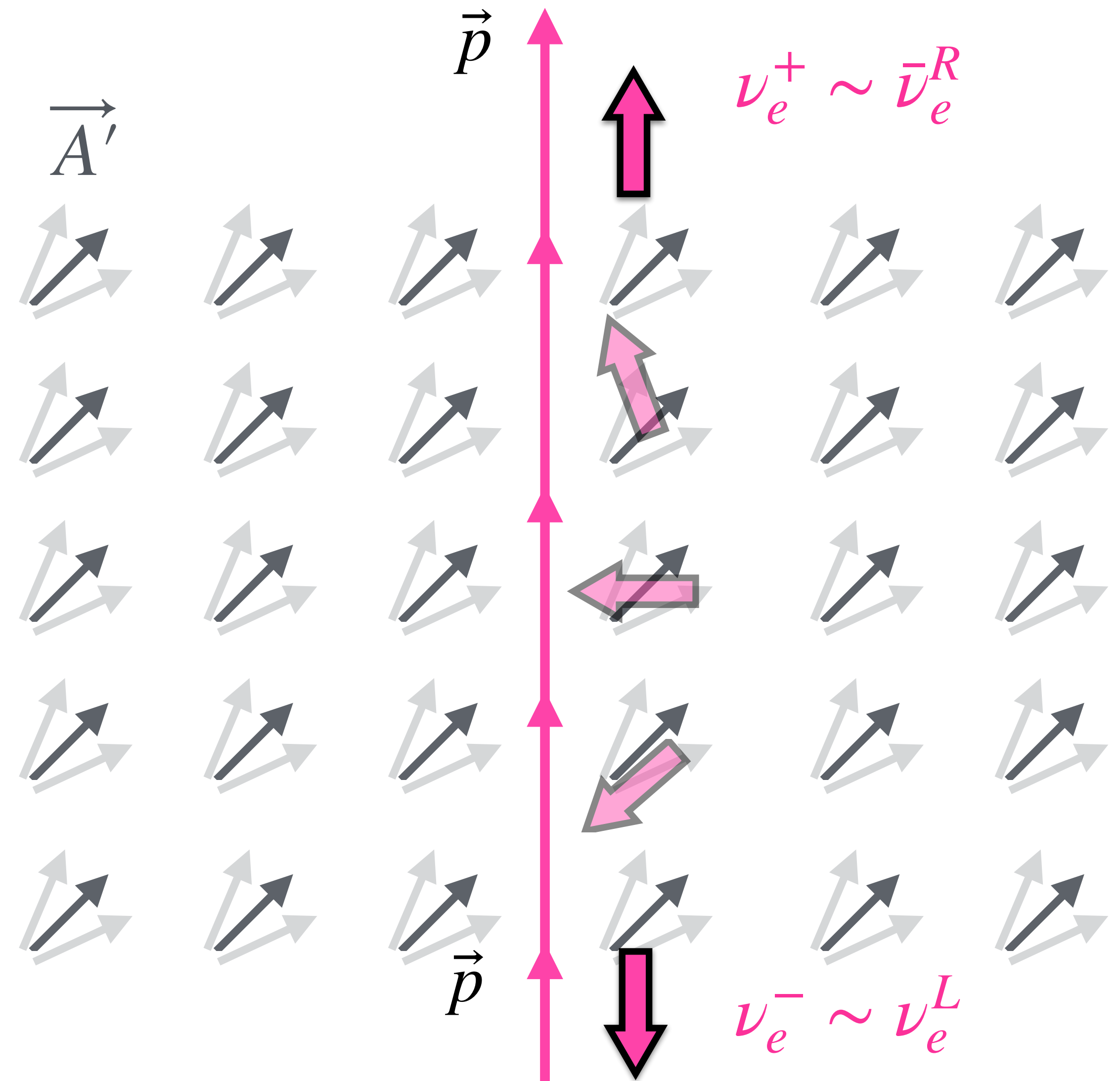
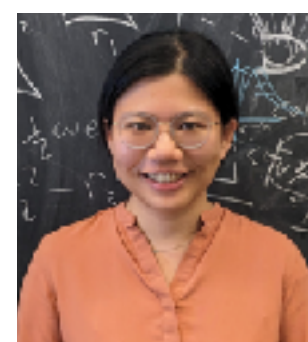
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In progress and in collaboration with:
A. Berlin, R. Capdevilla, T. Cheng, P. Machado (FNAL)



Neutrino “Rabi Oscillations”

The neutrino spin $\vec{s}$ precesses around the DM polarization vector $\vec{A}'$

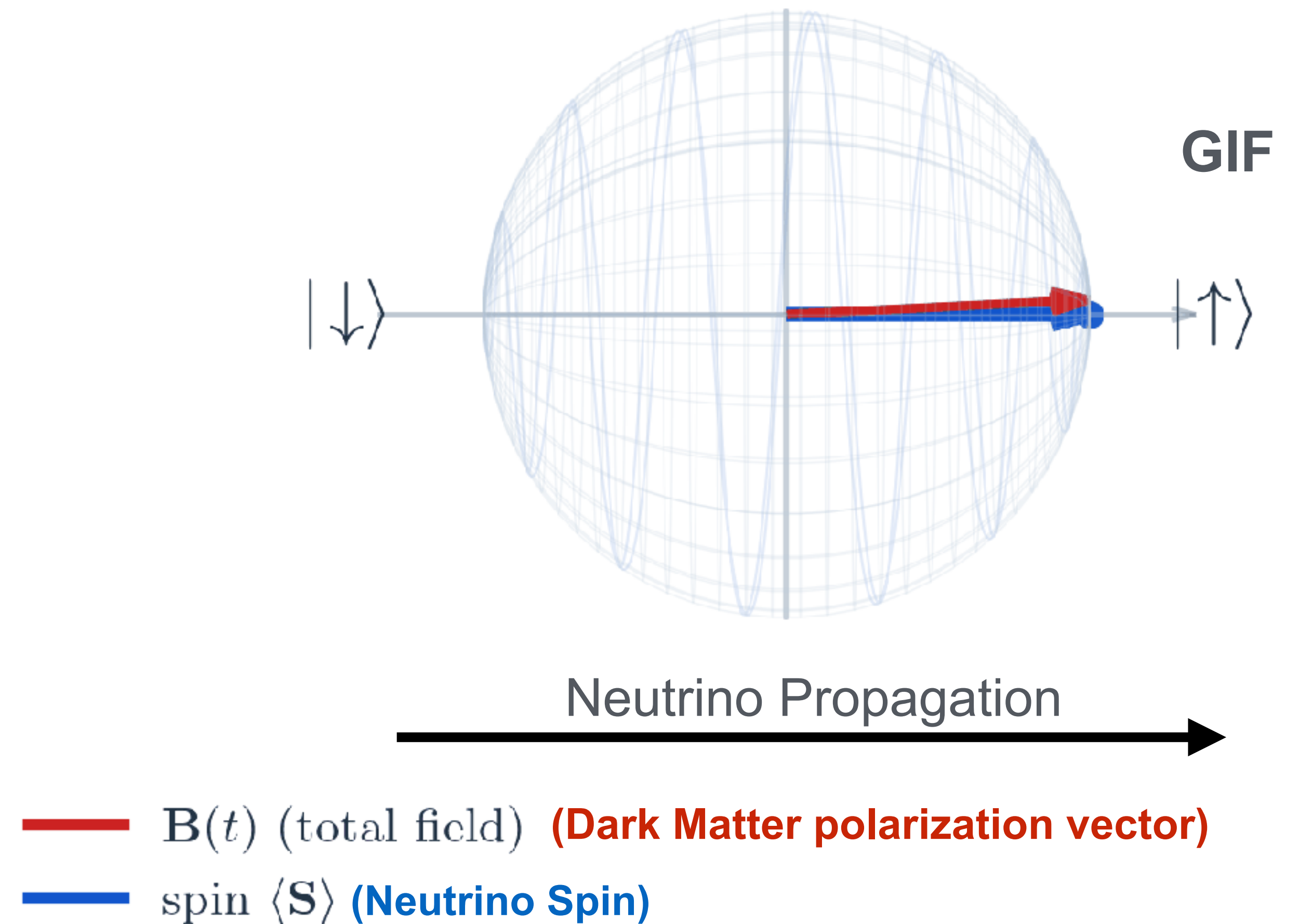
From the ν rest frame $A'_\perp \ll A'_\parallel$, so the field is mostly longitudinal.

Polarization of DM field is randomized at each spatial-coherent patch.

**For sufficiently long propagation time,
spin flip implies $\nu \rightarrow \bar{\nu}$ for
Majorana neutrinos.**

Note: For illustrative purposes only.

The neutrino spin flip is far slower!



Effective Hamiltonian of dark matter V interaction with the neutrino spin

$$H = V_{\parallel} \cos(\omega_0 t + \phi_z) \sigma_z + V_{\perp} (\cos(\omega_0 t + \phi_x) \sigma_x + \cos(\omega_0 t + \phi_y) \sigma_y)$$

Rabi oscillation with elliptically-polarized field

$$\omega_0 \sim \gamma m_{A'}, \quad V_{\parallel} \sim \gamma V_{\perp}, \quad V_{\perp} \sim e' \sqrt{2\rho_{\text{DM}}}/m_{A'} \quad \text{where. } \gamma = E_{\nu}/m_{\nu}$$

With the rotating wave approximation, the spin flip probability gives

$$P_{\nu \rightarrow \bar{\nu}} \simeq \sin^2(\Omega_{\text{eff}} t) \xrightarrow{\text{Small } t} P_{\nu \rightarrow \bar{\nu}} \sim \frac{e' \sqrt{2\rho_{\text{DM}}} L}{\gamma^2 m_{A'} v_{\text{DM}}} \times \varepsilon^2$$

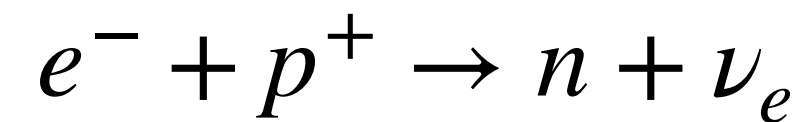
$$\varepsilon \equiv \sqrt{\sin^2(\phi_z - \phi_y) + \sin^2(\phi_z - \phi_x)}$$

“Ellipticity”

$\phi_{x,y,z}$ random phases in each coherent patch

Supernova $\nu \rightarrow \bar{\nu}$ conversion

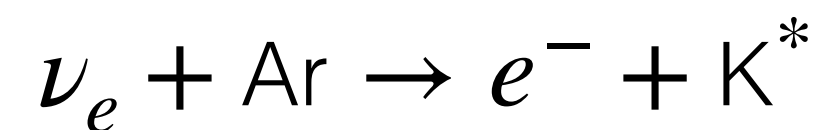
Huge ν_e burst from neutronization burst:



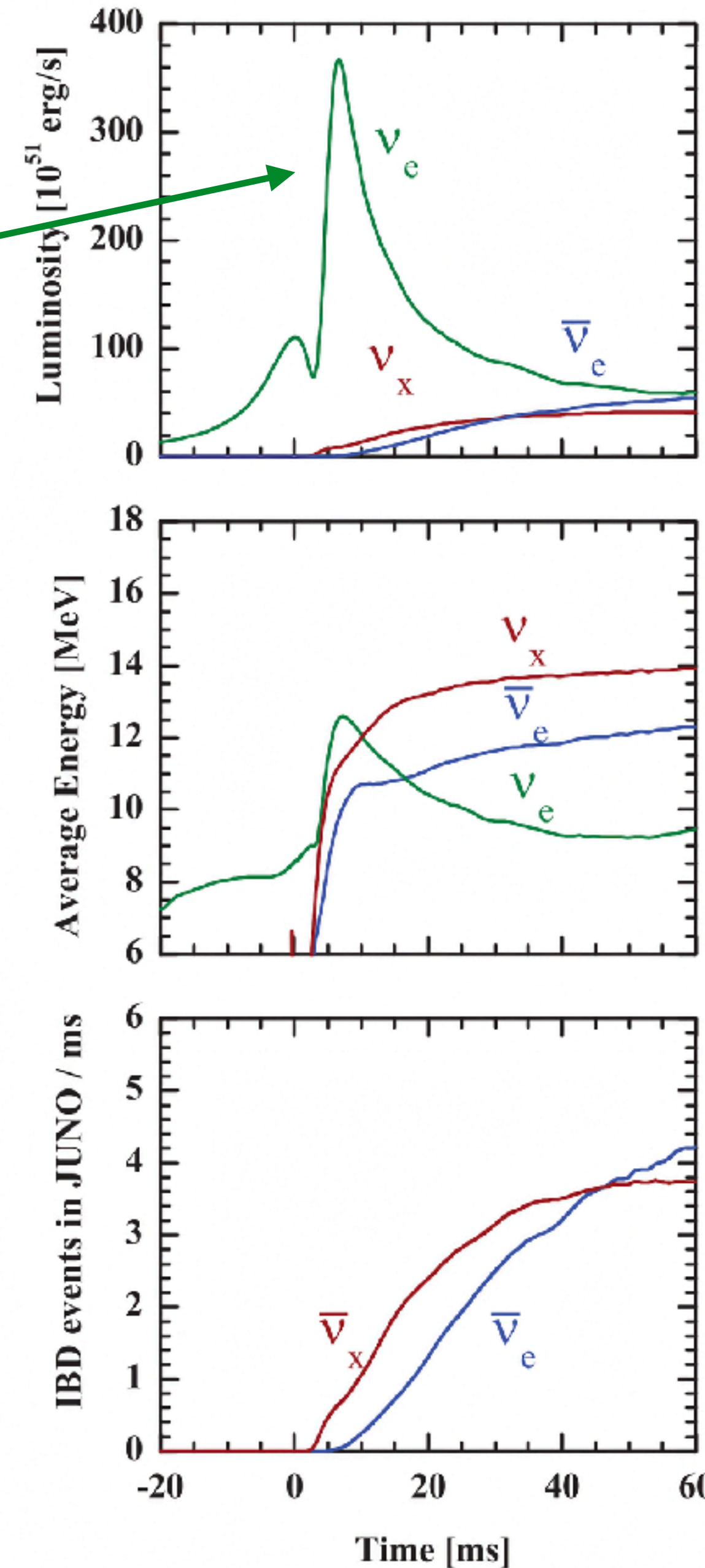
This feature is rather robust in supernovae:

$$F_{\nu_e} > F_{\bar{\nu}_e} \gtrsim F_{\nu_x} = F_{\bar{\nu}_x}$$

Especially interesting for **DUNE** that can see ν_e through



Neutronization $\mathcal{O}(30 \text{ ms})$

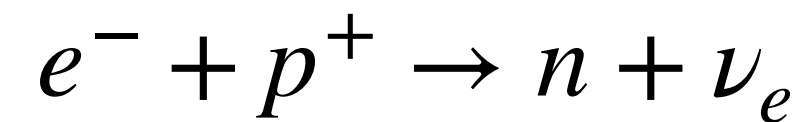


+ **Accretion phase**
 $\mathcal{O}(1 \text{ s})$ duration

Cooling phase
+ $\mathcal{O}(10 \text{ s})$ duration

Supernova $\nu \rightarrow \bar{\nu}$ conversion

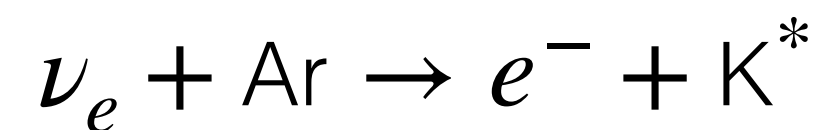
Huge ν_e burst from neutronization burst:



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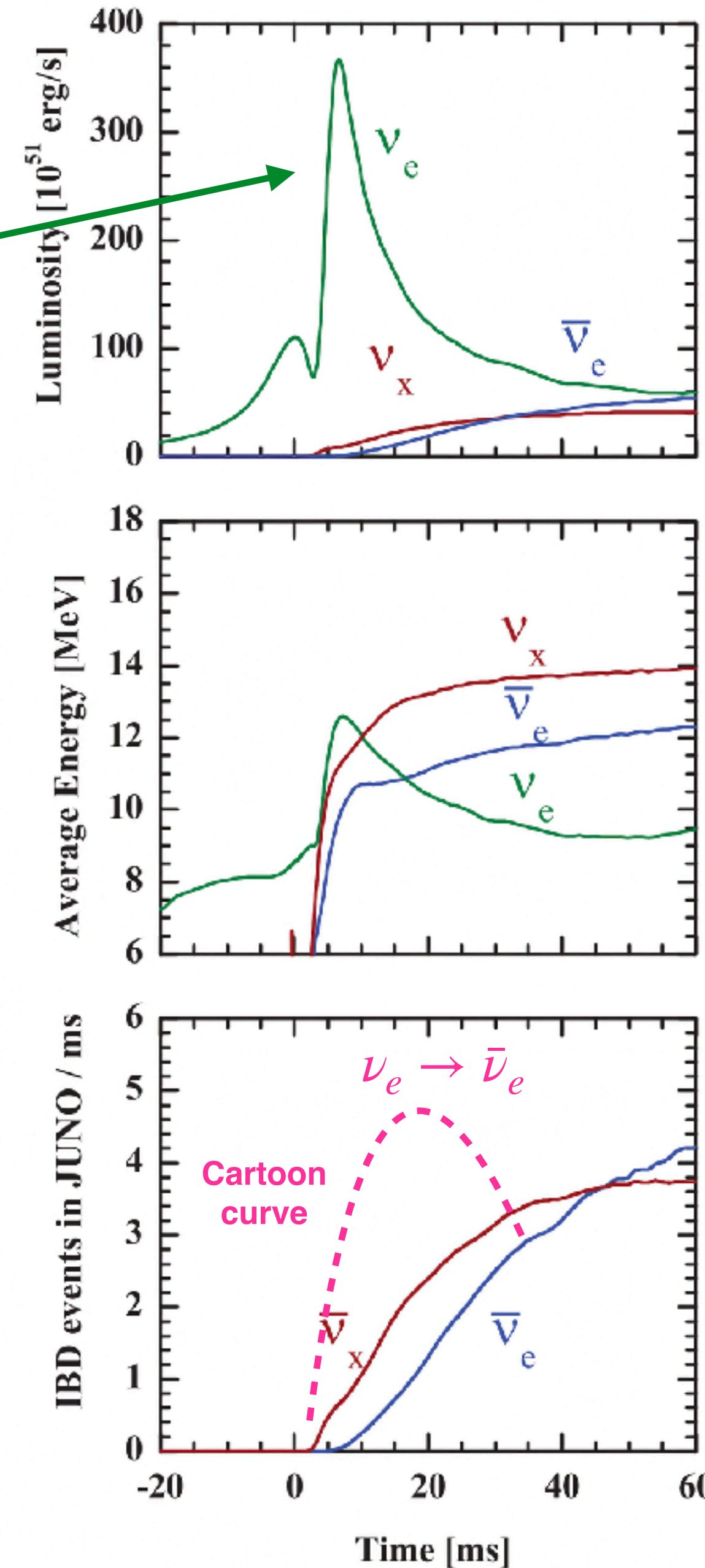
$$F_{\nu_e} > F_{\bar{\nu}_e} \gtrsim F_{\nu_x} = F_{\bar{\nu}_x}$$

Especially interesting for **DUNE** that can see ν_e through



The subsequent $\bar{\nu}_e$ components can be detected with IBD events at, e.g., **JUNO** and **Hyper-K**.

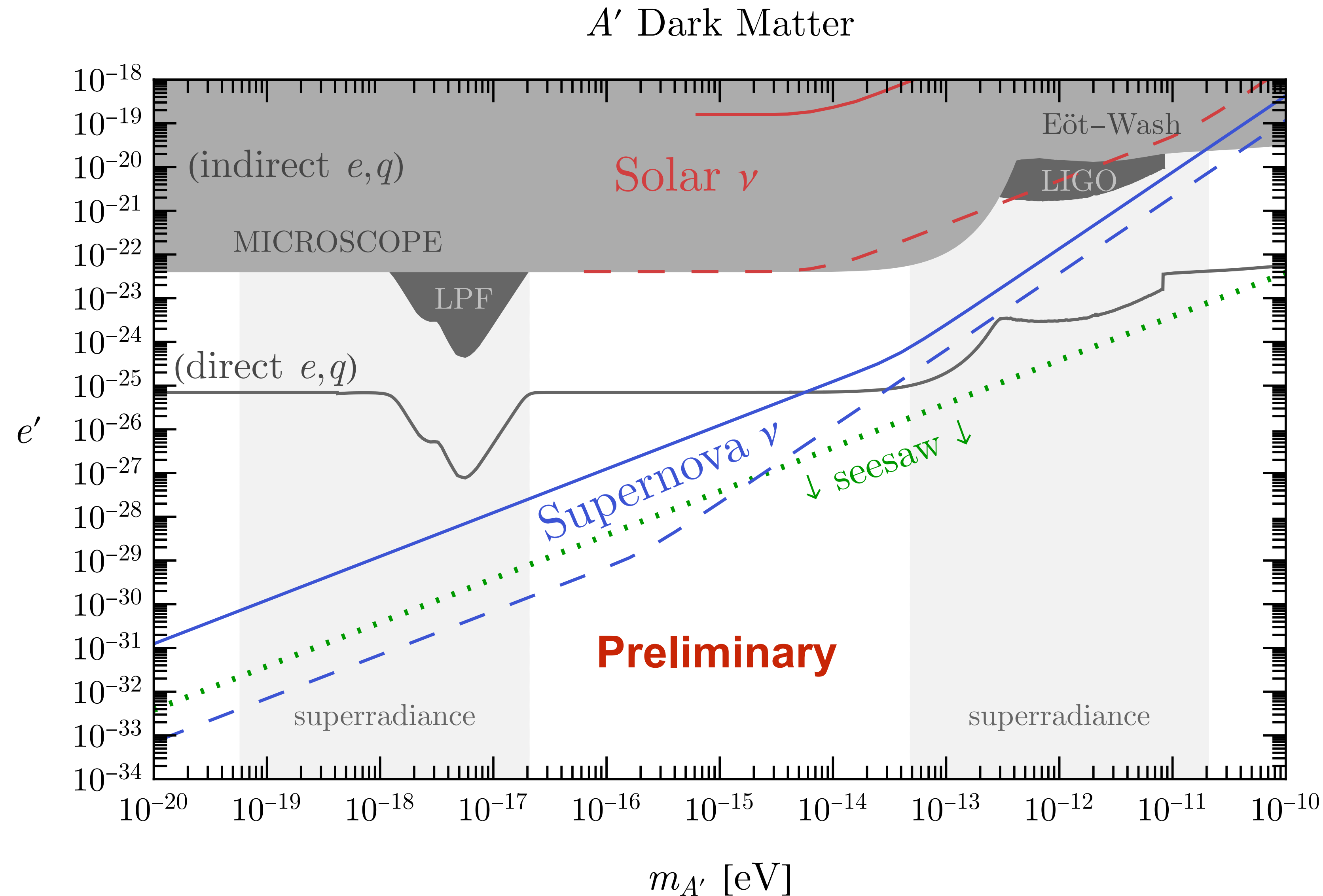
Neutronization $\mathcal{O}(30 \text{ ms})$



+ **Accretion phase**
 $\mathcal{O}(1 \text{ s})$ duration

Cooling phase
+ $\mathcal{O}(10 \text{ s})$ duration

Neutrino Spin Flip in the Vector Dark Matter Parameter Space



Regions of Interest

Vector dark matter
coupled to Majorana ν .

Dashed: largest ν mass
Solid: smallest ν mass

Equilibration
between ν and $\bar{\nu}$.

Take Home

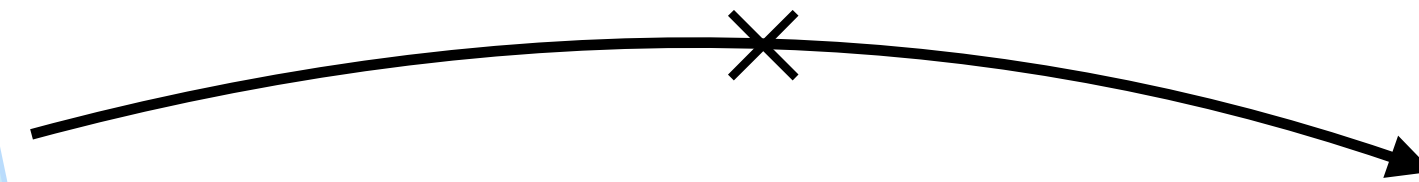
The SM may be incomplete already **at low energies.**
Neutrino masses are one crucial piece of the puzzle.

Right-handed neutrinos provide a unique opportunity in the lepton sector.
We are exploring it across multiple energy scales.

Neutrino physics is helping cast a wide net on new physics,
probing scenarios from **ultra-light to ultra-heavy dark matter**, new fermion
mass mechanisms, and a whole range of **lamppost dark sectors**.

STANDARD MODEL

$$(LH)N$$



Neutrino portal

$$F_{\mu\nu}X^{\mu\nu}$$



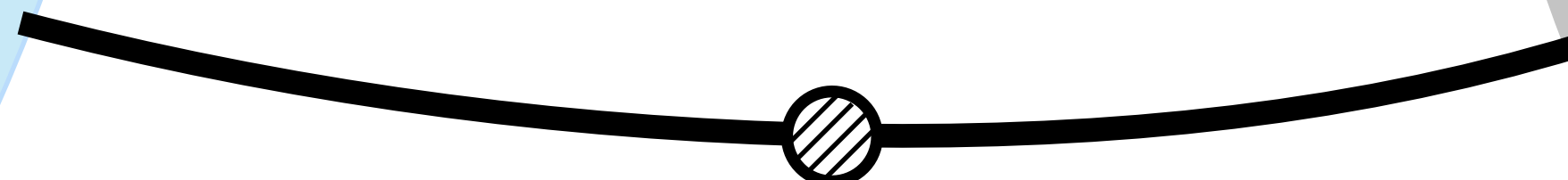
Vector portal

$$|H|^2|\phi|^2 + \phi|H|^2$$



Higgs portal

$$\mathcal{O}_{\text{SM}}\mathcal{O}_{\text{DS}}$$



Non-renormalizable portals

Context

Neutrino masses and mixing.
Seesaw partners.

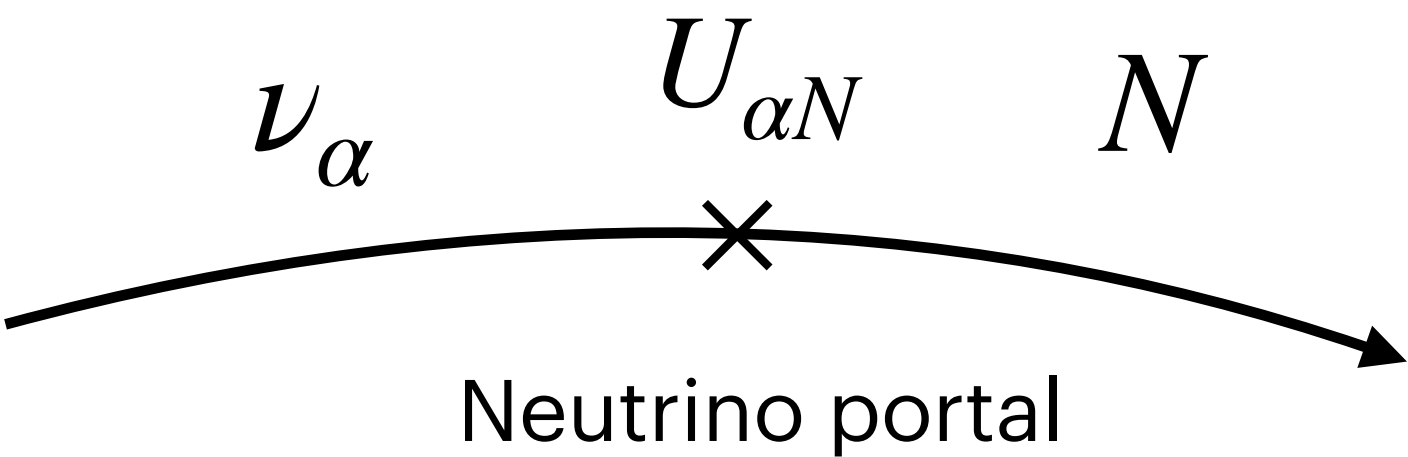
New $U(1)'$ gauge symmetry.
DM-SM force mediator.

Extremely general.
Scalar potential.

Hybrid approach:
light particles as messengers of UV scale.

STANDARD MODEL

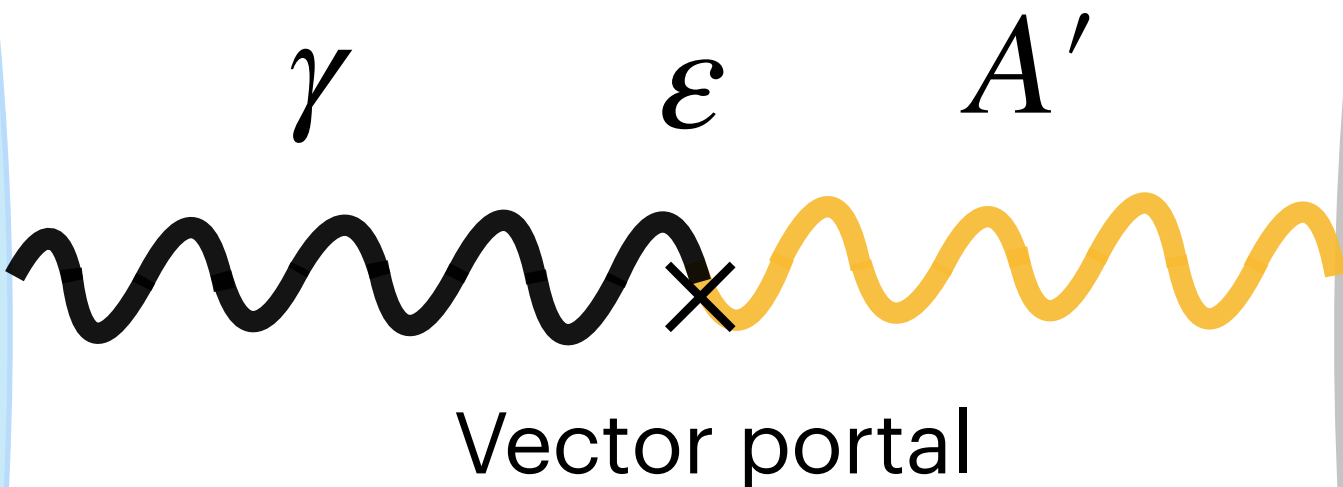
Decay width of dark particle to SM



Heavy Neutrinos

$$\Gamma \sim |U_{\alpha N}|^2 G_F^2 M_N^5$$

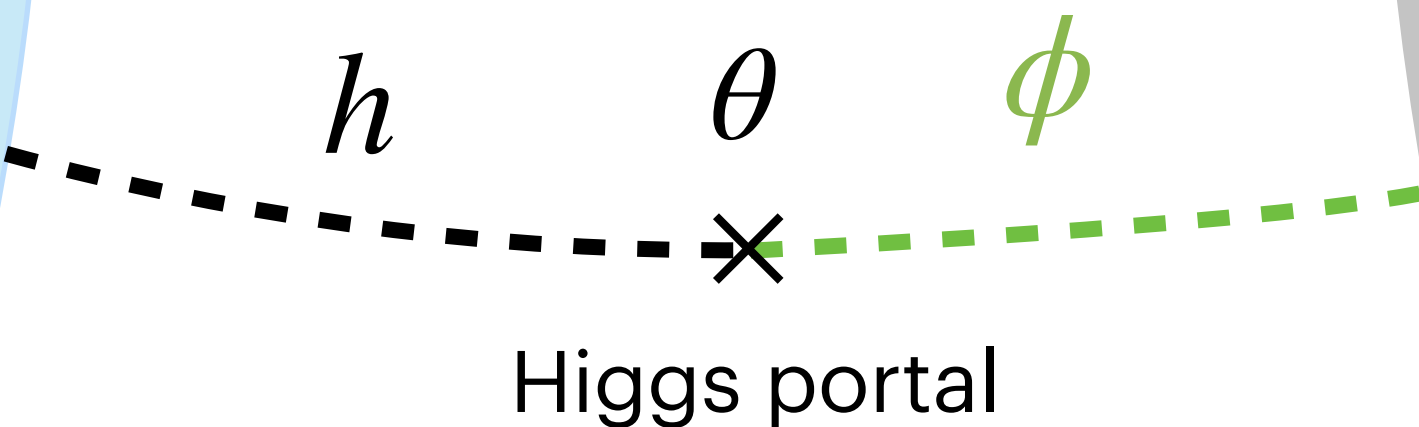
weaker-than-Weak



Dark photons

$$\Gamma \sim (e\epsilon)^2 m_{A'}$$

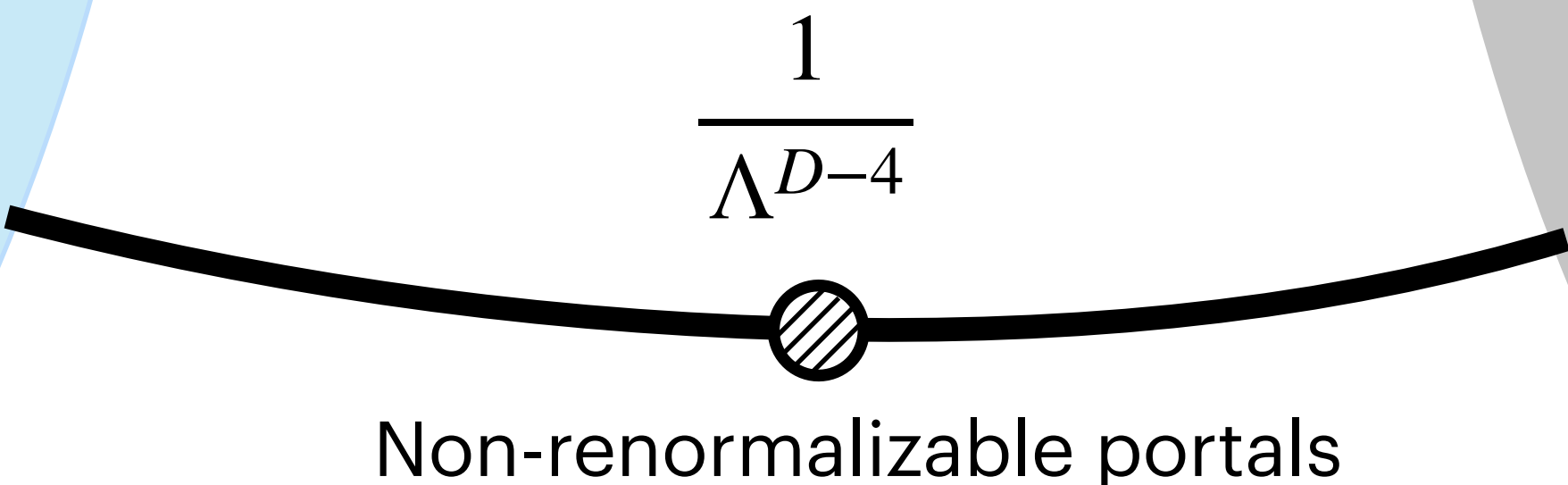
weaker-than-EM



Dark Higgs

$$\Gamma \sim \theta^2 y_f^2 m_S$$

weaker-than-Yukawa



Axion-like-particles,
 χ SMEFT, ν SMEFT, etc.

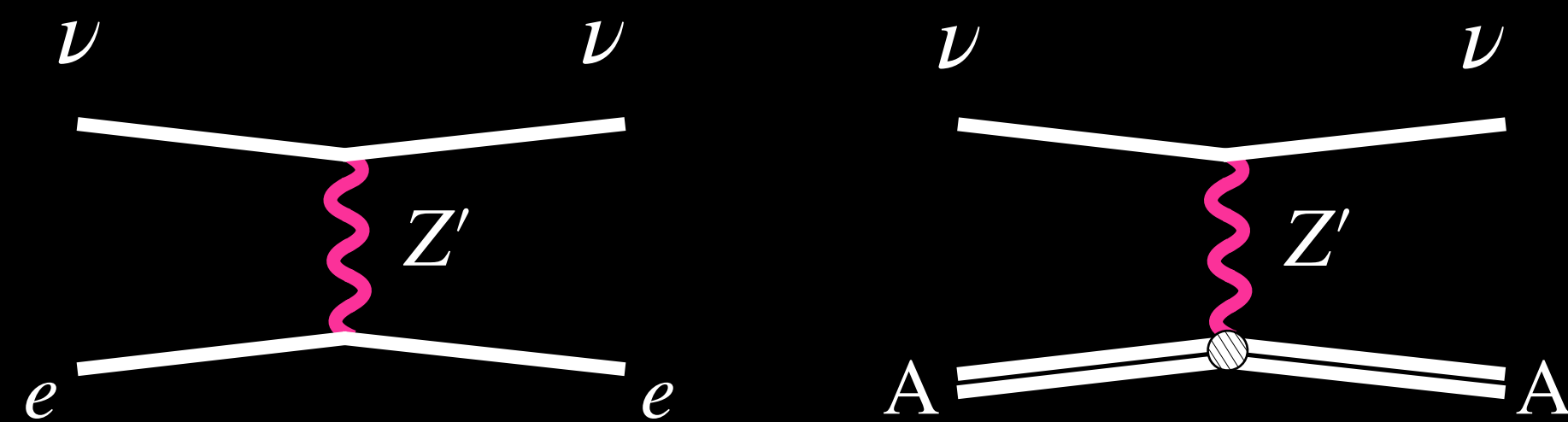
$$\Gamma \sim \frac{m_a m_e^2}{\Lambda^2}$$

e.g. ALP decay constant Λ

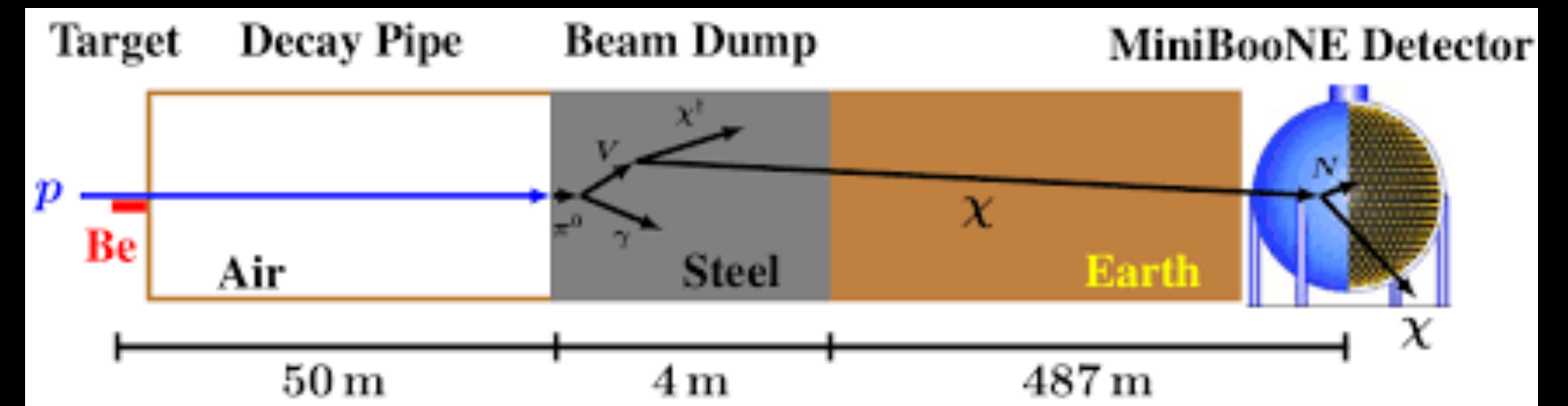
Other examples of beyond-the-Standard Model applications:

New Forces

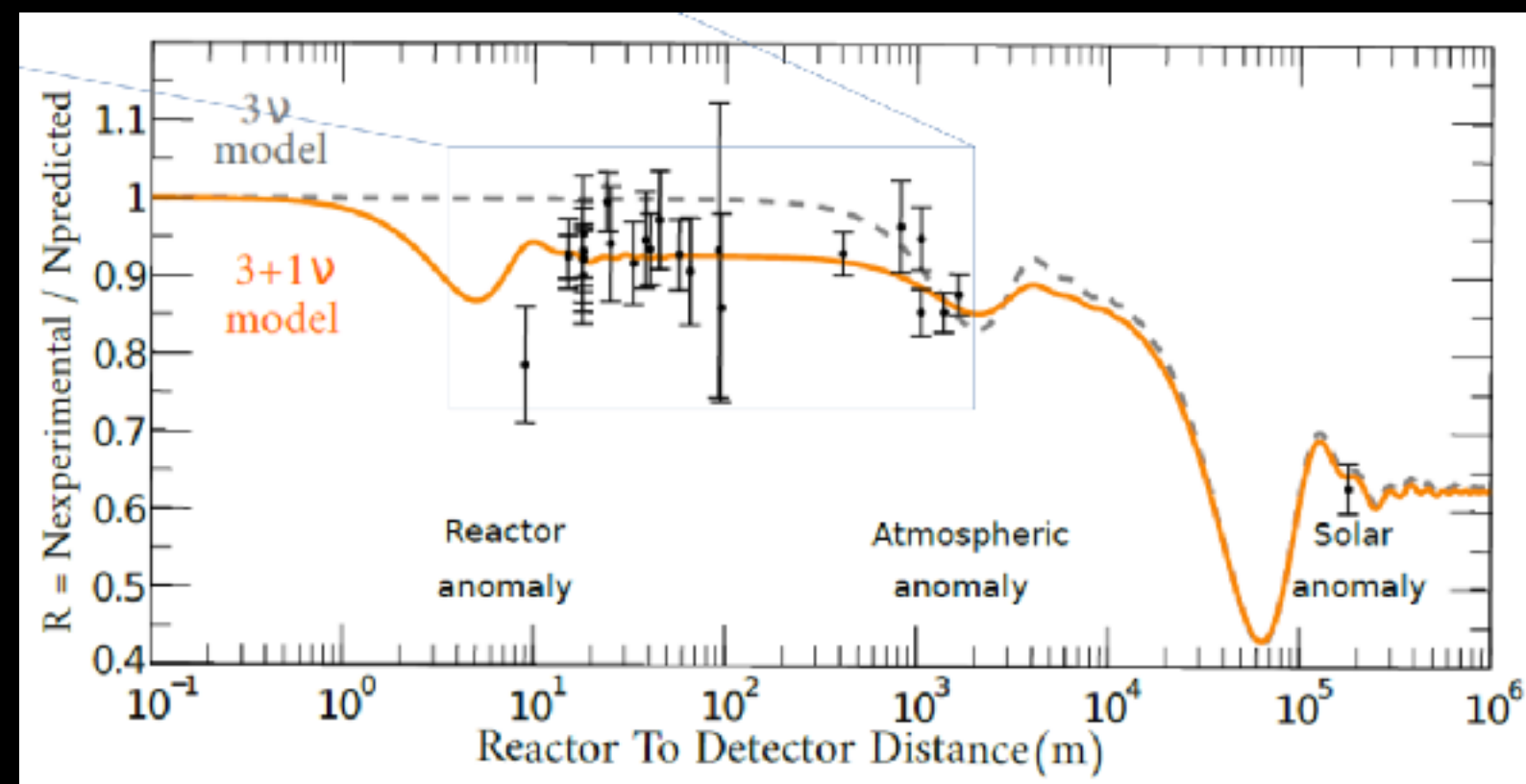
Non-Standard Interactions (NSIs)



Light Dark Matter



Sterile Neutrino Oscillations



Long-Lived Particles

